Empirical Site Separation: Bending, Tearing, Contextuality and Holonomy in a National Survey, Retrained Language Models and Clinical Data

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Abstract

Language models are retrained, survey questions are reworded, clinical populations drift, and in each case the question is the same: when the information changes, did the structure only bend, did it tear, did it break, and what changed that the summary numbers do not show? We turn the programme's theory into a monitor and run it on GPT-3.5 and GPT-4 before and after their June 2023 update (eight question sets, including medical licensing questions asked with intact and with damaged prompts), on the General Social Survey, 1985–2024, on two cycles of NHANES, and, in pre-registered analyses with replication samples, on intensive-care, hospital and cardiology records and on two COVID-19 mental-health panels of up to six waves. On the survey we test the separation of sites of the programme: that contextuality (Layer 3), the geometry of couplings (Layer 2) and the holonomy of operations (Layer 1) are different invariants. On 1320 system-years of eleven spending items in two randomized wordings, a contextuality test that assumes identical connections reports a contextual fraction of 0.02–0.58 in every one, while Contextuality-by-Default in canonical form absorbs all of it as direct influence: no item pair is detected as contextual after combining its years and correcting for the 55 pairs (the closest, crime and drugs, has $p = 0.0073$ against a threshold of 0.0009). The same wordings carry a non-flat connection, a nonzero holonomy class in $H^1 = \mathbb{R}^{10}$ in every one of the 24 years, dominated by "welfare" against "assistance to the poor"; and the items are often dependent, a nonzero distance from the entropy brane's intersection point (1140 of 2640 tables at a Bonferroni level). The three invariants dissociate on the same systems, as the stratification of the programme requires, and no one scalar orders them. We draw the coupling polytopes of real margins as moment polytopes of toric surfaces with their nonzero torus branes: the maximum-entropy coupling never carries one and the leximin coupling always carries at least half of them, and the one decided survey tear is a toric blow-down at a fixed point, taking the 3×3 polytope from 15 to 12 vertices. Retraining tears are rank reversals of the two models; damaged prompts flip the correctness of 10–17% of items while moving accuracy by 1–5 points. In the survey panels the answers of three waves do not compose, but a latent Markov chain seen through unreliable answers is not rejected in any panel, and the order of a respondent's answers adds nothing to their association with the 2008 vote or party identification beyond which answers were given. A published behavioural experiment serves as a positive control: the programme's cyclic criterion

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[*] Code and checks: `https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/tree/main/monitor`. `check_monitor.py` recomputes or verifies every number reported here. All papers of the programme: `https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/tree/main/papers`.

reproduces its contextuality exactly; a planted-data check of the canonical test used elsewhere shows that it holds its size but detects only large contextuality at the sample sizes here, so its non-detections rule out large effects only. In pre-registered analyses of open clinical and panel data, available-case tables of a selectively measured quantity are incompatible beyond sampling (lactate in two independent ICU samples, HbA1c in 101,766 diabetic encounters, attrition in the UK panel): measurement is not at random with respect to the measured states. With four waves, a latent first-order Markov chain is not rejected in either COVID-19 panel, although the answers do not compose: composition failure in the answers is consistent with measurement error on a Markov latent state, though at these sample sizes the test has limited power against longer memory. No category broke at the 1% level. The algebraic backbone, which no public data can reach, is tested on synthetic data with closed-form ground truth and planted failures. Kleisli composition, the three-level tower, its mixed distributive law and the interchange of couplings with kernels behave exactly as stated, and every predicted failure appears at its predicted place with its predicted size. The one exception is instructive: a random-instance check is blind to a planted failure. On real panels one theorem of the backbone, the interchange law, shows that banding inflates the composition failure: the monitor's null is 41–51% too small, although the failure survives the correction. A three-level tower fitted on PhysioNet is idle at its reliability level between random halves of one population, as it should be. Under a shift between units it cannot represent the shift as specified, and its own lifting certificate measures the gap. Its Fano tubes, the toric types of the predicted system, show this as a confidently wrong chamber. With a family of floors, a posterior over binary channels, the tower's reliability level earns its keep in every pre-registered shift between ICU unit types and harms nowhere. The geometry of GSS item pairs forecasts their next chamber change beyond the margins, while per-stay ICU systems are almost all broken and their geometry forecasts nothing. Selection models not built on the triangle cells fit the UK panel's attrition and fail for ICU lactate and diabetic HbA1c. The map from information to outcome is tested for context invariance as membership in a toric variety. Four of eleven GSS items, and ICU unit types, leave it, and the toric type of the answer × outcome system records where. Point polytopes at the boundary turn out to be mostly artefacts of zero counts: their geometry is recoverable from a posterior, but it does not forecast death. A tracker built on the family of coupling polytopes and their normal fans shows the approach to a wall before the one decided survey tear and in the physiology of ICU patients who die. In recursive training with known truth, the clearest sign of destroyed information is the contexts' steady departure from the invariance variety, which fresh data cap; three of five pre-registered predictions fail, and we report them. A decision layer on the same geometry turns sources of known reliability into auditable decisions, with a support threshold, the weakest link and a typed record of why a conclusion fails to glue. Support floors on cells preserve the toric geometry exactly, by a shift to reduced margins, and a new obstruction appears: families that glue, but not with the floor. The ambience is carried by the total category of a functor from contexts into posets: policies act as an idempotent comonad on each fibre, summarising by floors as an idempotent monad, and their failure to commute with transport is a computable oplax 2-cell. On real scientific claims (SciFact) with weak verifiers, the layer is right when it acts and its wrong acts stay within a conformal guarantee, but it rarely acts, and its margin ranks pairs worse than plain confidence. We explain why the other deeper layers are not yet activated by the data. We state where the programme falls short and list every dataset used across its ten papers.

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1 The question

When a model is retrained, a question is reworded or a population changes, what is usually reported is a summary: an accuracy, a share of respondents, a prevalence. The programme of which this paper is part [7, 8, 9, 5] claims that summaries miss the structure of the change. Two quantities measured on the same units define a coupling, the coupling lies in a polytope fixed by the margins, and the polytope changes in three ways as the margins move: it bends within a chamber, it tears across a wall where it changes combinatorial type, and it breaks at the boundary where a margin reaches zero [7, Thm. 3.3]. Over several contexts, the same content measured differently may be contextual or merely directly influenced [8, §4]. And when the same units are seen twice, the coupling of a content with itself across contexts can carry change the margins do not show.

This paper tests these claims on data where the information demonstrably changed:

- *retraining*: GPT-3.5 and GPT-4, March versus June 2023, on the same questions [10], and, for the medical questions, intact versus damaged prompts;
- *society*: the General Social Survey, where the wording of spending questions is randomized and where three-wave panels follow the same people [19, 29];
- *clinical*: two NHANES cycles, the medical licensing questions inside the retraining data, and, under plans written before the data were opened, two independent ICU samples, 101,766 diabetic encounters and four heart-disease sites;
- *longitudinal*: the UK and Irish COVID-19 mental-health panels, with up to six waves on the same people;
- *a positive control*: a published behavioural experiment in which contextuality is known to be present;
- *the structural backbone*: synthetic data with closed-form ground truth and planted failures, for the monad of kernels, the three-level tower, its distributive law and the interchange law (section 10).

None of these designs identifies the effect of a policy; the survey analyses describe how measured attitudes change, not why.

The survey also tests a structural claim of the programme. ThreeLayers [33] organizes the theory in three layers, each with its own invariant on its own site: the holonomy of operations (Layer 1), information on a site with a global joint (Layer 2), and contextuality on a cover without one (Layer 3); each is nontrivial on its own site and zero on the others [33, §2], and for the coefficient systems of Layers 2 and 3 this is made precise by the non-unification and vanishing theorems [33, Thms. 15.1, 15.3]. These are structural statements, not predictions about data. Its randomized wordings, its forty years of cross-sections and its panels let us compute all three on the same respondents (section 6), after checking on a published experiment that the test detects contextuality where it exists (section 6.2). Sections 7 and 8 test the remaining claims, with plans fixed in advance and replication samples, on clinical records and on six-wave panels. Section 9 then draws the toric and Floer-theoretic picture of [8] for margins taken from these data, and section 18 lists the data of the whole programme.

2 The monitor

Definition 2.1 (System)**.** A *system* is a pair of contents A, B with finitely many outcomes, measured jointly on the same units in each of several contexts. In context c the joint table T_c has margins (f_c, g_c). Contexts are versions of a model, prompts, wordings of a question, survey years, panel waves or survey cycles.

Bands. Each margin is given an elementwise band: Wilson intervals at level α/k for k outcomes, $\alpha = 0.05$, then tightened so that the band is consistent with total mass one [7, Prop. 2.2]. For weighted surveys the sample size is Kish's effective size $(\sum w)^2/\sum w^2$ [22]; clustering and stratification are not modelled.

Regime. The walls of a pair of margins are the hyperplanes $f(I) = g(J)$; the tear margin m_T is the distance to the nearest wall and the break margin the smallest margin [7, Thm. 3.2]. A wall is *decided* on a band when the range of $f(I) - g(J)$ over the band excludes zero [7, Prop. 3.4]. A change $c \to c'$ is a

- *tear* if some wall decided in both states changes side;
- *possible tear* (tear?) if the point estimates change side of a wall that is undecided in at least one of the two states;
- *break* if an outcome whose band lies above 1% in c has its band below 1% in c';
- *bend* otherwise.

The *stress* of the change is $s = \big(\mathrm{TV}(f_c, f_{c'}) + \mathrm{TV}(g_c, g_{c'})\big)/m_T(f_c, g_c)$.

Proposition 2.2 (Tears need stress)**.** *If the point estimates of a change cross a wall, then $s \geq 1$.*

Proof. The segment from (f_c, g_c) to $(f_{c'}, g_{c'})$ meets the wall, every point of the wall is at distance at least $m_T(f_c, g_c)$ from (f_c, g_c) in the metric $\mathrm{TV} + \mathrm{TV}$ [7, Thm. 3.2], and the distance from (f_c, g_c) to the crossing point is at most the length of the segment, which is the realized change. $\square$

So stress below one certifies a bend; stress above one is necessary but not sufficient for a tear. Since every tear has stress at least one by the proposition, the stresses reported below carry information only through their size.

Remark 2.3 (Near-empty categories). When an outcome is nearly empty in both contents (a model almost never fails to answer), the wall comparing the two small shares is very close, m_T is tiny and the stress is large even for a small change. Such walls are never decided on the bands, so they produce possible tears, not tears; we flag them where they occur.

Context. We test Contextuality-by-Default in its canonical form [17]: each content is replaced by all its dichotomizations (for three outcomes, each outcome against the other two), and the system is noncontextual exactly when some coupling of (A_1, B_1, A_2, B_2) reproduces the two context tables and couples every dichotomization of A, and of B, maximally with itself across the two contexts. This is a linear program; its *deficit*, the maximal minus the attainable total agreement of the split connections, is positive exactly when the system is contextual. For binary contents it reduces to the cyclic system of rank 2 [16, 24], contextual exactly when $|\langle A_1B_1\rangle - \langle A_2B_2\rangle| > \Delta$ with $\Delta = |\langle A_1\rangle - \langle A_2\rangle| + |\langle B_1\rangle - \langle B_2\rangle|$ the direct influence of the context on the contents; we also report this rank-2 statistic for the dichotomization "first outcome against the rest", computed both by the criterion and by the linear program of [8, §4], which agree on every case. A three-outcome system can be contextual in its splits while this single dichotomization is not, and the code's unit tests show it happens often (219 of 400 random systems that are noncontextual in the single dichotomization are contextual in the canonical form).

Churn. When the same units are observed in two contexts, the coupling of a content with itself across them is observed, not assumed. Its agreement $P(X = Y)$ is at most $\sum_i \min(f_i, g_i)$, the agreement of the maximal coupling that Contextuality-by-Default assigns to a connection.

Proposition 2.4 (Excess churn and masked change)**.** *The* excess churn $\sum_i \min(f_i, g_i) - P(X = Y)$ *is nonnegative, and zero exactly when the observed coupling is a maximal coupling. For a binary outcome with* 1 *meaning correct, let* $\ell = P(X = 1, Y = 0)$ *be the mass lost and* $\gamma = P(X = 0, Y = 1)$ *the mass gained, both as shares of all units. Then the excess churn is* $2\min(\ell, \gamma)$*: twice the* masked change*, the losses that are offset by gains and so invisible in the net change* $\gamma - \ell$*.*

Proof. A coupling puts at most $\min(f_i, g_i)$ on each diagonal cell, with equality on all of them exactly for a maximal coupling. For two outcomes, $\sum_i \min(f_i, g_i) = 1 - |\gamma - \ell|$ and $P(X = Y) = 1 - \ell - \gamma$, whose difference is $\ell + \gamma - |\gamma - \ell| = 2\min(\ell, \gamma)$. ☐

The lost mass ℓ is what the model-update literature calls the rate of *negative flips* [38]; the proposition says that the part of it that the net accuracy hides is exactly half the distance of the observed coupling from the maximal one. Excess churn mixes real change with measurement noise: a model sampled at positive temperature, or a respondent answering a survey twice, disagrees with itself. It is an upper bound on masked change until a noise floor is subtracted, and none of the data here contain true replicates.

Composition. For three waves on the same units, the gluing gap $I(X_1; X_3 \mid X_2)$ vanishes exactly when the joint law of the three waves is the gluing of the first two waves with the last two [7, Prop. 4.8], that is, when the answers form a Markov chain. A positive gap does *not* by itself mean that the underlying change fails to compose: a stable attitude measured with independent error at each wave also gives a positive gap [15, 2, 37]. We therefore test the observed answers against the glued (Markov) law, by resampling from it, and fit three latent-class models with three classes, by EM from 30 starts: *stable* (a fixed latent class answered with the same response law at every wave; 8 parameters), *drifting* (a fixed class with a response law for each wave; 20 parameters) and *latent Markov* (the class follows a Markov chain over the waves, answered with one response law; 20 parameters) [30]. Each is tested by the likelihood ratio G^2 against the saturated table, on 18, 6 and 6 degrees of freedom. The latent Markov model is the gluing of the programme one level down: if it fits, the attitudes may compose while the answers do not. The gap is a plug-in estimate, biased upward by about 0.014 bits at $n \approx 600$; the Markov null carries the same bias.

3 Data

Retraining. The LLM-drift data of Chen, Zaharia and Zou [10] contain the answers of GPT-3.5 (versions 0301 and 0613) and GPT-4 (0314 and 0613) to the same questions in eight tasks, at temperature 0.1. Each answer is coded *correct*, *wrong* or *no valid answer*, using the last stated answer ("the answer is (X)", the last bracketed yes or no, the last boxed number). For the sensitive questions the outcome is answered or refused, for LeetCode directly usable code or not, and for OpinionQA, which has no correct answer, answered (a valid option) or not. The USMLE file asks each of its 340 questions twice: once with the intact prompt and once with a damaged one, in which a stretch of text is deleted: a passage of the question stem in 89% of the questions, option (A) in 10%; two pairs are identical. We use the intact prompts for the update and treat intact versus damaged as a second change of information. Pooling both prompts reproduces the published GPT-4 accuracy, 86.6% → 82.1%; our figures for primes (84.0% → 51.1%) and for its response rate to sensitive questions (21% → 5%) also match. A task's system has contents A = GPT-3.5 and B = GPT-4, measured on the same items, and contexts March and June.

Society. The General Social Survey [19], read through the `gssr` package [20], asks whether the government spends too little, about the right amount or too much on a list of items, each in one of two randomly assigned wordings: X ("welfare", "solving the problems of the big cities", "improving the conditions of Blacks", . . .) and Y ("assistance to the poor", "assistance to big cities", "assistance to blacks", . . .) [29, 21]. No respondent gets both wordings of an item. We use three pairs of items, 1985–2024, with the weight `wtssps`, and in section 6 all eleven items asked in both wordings. A system has two items as contents and the wordings, or the years, as contexts. The 2006–2010, 2008–2012 and 2010–2014 panels follow the same respondents through three waves and record their 2008 presidential vote; we use the respondents who answered the welfare item in the same wording in all three waves, unweighted.

Clinical. NHANES 2009–2010 and 2011–2012 [27], adults 20+ with exam weights: contents diabetes (self-reported diagnosis) and BMI class (below 25, 25–30, 30 and above); contexts the two cycles. The medical licensing questions (USMLE) in the retraining data are the second clinical system. Open, de-identified hospital data are used in section 7; the credentialed electronic health records of MIMIC-IV [26] are not, because their data use agreement does not allow them to leave the credentialed user's machine. The protocol of [7, §11] for them is provided as a kit to be run locally.

Open clinical and longitudinal data. PhysioNet 2012 ICU stays (sets A and B) [28, 18], the UCI Diabetes 130-US Hospitals [32, 35] and Heart Disease [12, 36] data, and the UK and Irish strands of the C19PRC panels [25, 31] are described with their analyses in sections 7 and 8; the Snow Queen tables [14] in section 6.2; the synthetic data of the structural tests in section 10.

4 Retraining

Tears are reversals. Three changes are tears decided on the bands: primes, happy numbers and HotpotQA, and in all three GPT-3.5 and GPT-4 trade places. On primes and happy numbers the wall crossed first is $f_{\text{correct}} = g_{\text{correct}}$: GPT-3.5 rises from 52.9% to 76.0% on primes while GPT-4 falls from 84.0% to 51.1%. On HotpotQA the order reverses the other way. The sensitive questions give a possible tear, with stress 1.2; there too the two models trade places (GPT-3.5 answers

Table 1: GPT-3.5 and GPT-4, March → June 2023 (USMLE: intact prompts). Accuracy (OpinionQA, sensitive questions: response rate) in percent. Regime and stress of the system (GPT-3.5, GPT-4) from March to June. s/Δ: the rank-2 Contextuality-by-Default statistic for correct against not, against the direct influences; the canonical test (all dichotomizations of the three outcomes) is noncontextual in every task as well. Lost/masked: percent of all items correct in March and not in June, and the part of it offset by gains (proposition 2.4).

task	items	GPT-3.5	GPT-4	regime	stress	s/Δ	lost/masked	
							GPT-3.5	GPT-4
USMLE (medical)	340	63.5→65.0	88.2→84.7	tear?	19.0	0.01/0.10	7.6/7.6	4.7/1.2
OpinionQA	1506	100.0→100.0	97.9→24.7	bend	35.6	1.46/1.46	0.0/0.0	73.3/0.1
sensitive questions	100	2.0→8.0	21.0→5.0	tear?	1.2	0.28/0.44	0.0/0.0	16.0/0.0
primes	1000	52.9→76.0	84.0→51.1	tear	152.8	0.00/1.12	17.3/17.3	47.6/14.7
happy numbers	500	30.6→48.2	83.6→35.2	tear	25.8	0.22/1.32	14.0/14.0	53.8/5.4
ARC	467	10.9→14.1	24.4→27.0	bend	27.0	0.04/0.12	1.7/1.7	3.2/3.2
LeetCode (easy)	50	22.0→2.0	52.0→10.0	bend	2.4	0.52/1.24	20.0/0.0	46.0/4.0
HotpotQA (agent)	7405	22.3→13.8	1.2→37.0	tear	12.7	0.11/0.89	12.7/4.2	0.4/0.4

2% → 8%, GPT-4 21% → 5%), but 100 questions do not decide the wall. USMLE is a possible tear of the kind of remark 2.3: the wall crossed compares the two models' shares of invalid answers, both below 1%, and the stress of 19 reflects that wall's closeness, not the size of the change.

A degenerate system. OpinionQA is the largest change in the data, GPT-4's response rate falling from 97.9% to 24.7%, and the monitor calls it a bend. That is correct and instructive: GPT-3.5 answers every question in both months, so its margin is a vertex, the coupling polytope is a single point in both months, and no change of GPT-4's margin can tear it. A system with one content at a vertex has no structure to tear; the change must be read from the other content's margin.

The change is direct influence, not contextuality. In every task $s \leq \Delta$, and the canonical test finds no contextuality either. Apart from OpinionQA, where GPT-3.5 is constant and $s = \Delta$ identically, the margins $\Delta - s$ range from 0.08 (ARC) to 1.1 (primes and happy numbers). The update changed each model's distribution of answers, and that direct influence accounts for the joint tables of the two models.

Masked change. On intact USMLE prompts GPT-4 lost 4.7% of all items (5.3% of the items it had right in March) and regained 1.2%, so its net drop of 3.5 points hides little; GPT-3.5's gains (9.1%) more than offset its losses (7.6%), all of them masked. On primes GPT-3.5 improved by 23 points and still lost 17.3% of all items, a third of its March correct answers. Without replicate answers these are upper bounds on masked damage: decoding noise at temperature 0.1 cannot be separated from them.

Damaged information. Table 2 changes the information instead of the model. Damaging the prompt costs between 1.1 and 5.3 points of accuracy, but it changes the correctness of 10–17% of the questions: most of the item-level change is masked in the accuracy. Both models lose more to the damaged prompt in June than in March (GPT-4 3.2 → 5.3 points, GPT-3.5 1.1 → 4.4), which would mean that the update made them more fragile to damaged information; but the paired intervals include zero, so these data do not establish it.

Breaking. No outcome fell below the 1% band, so no change is a break at that level. The nearest case is GPT-4's "no valid answer" on happy numbers, 9.4% in March and exactly 0 in June: the category emptied, but with 500 items the upper band is 1.1%.

Table 2: USMLE, intact versus damaged prompt, same 340 questions; differences are computed before rounding. Accuracy in percent; lost and gained: percent of all questions whose correctness changes from intact to damaged; excess churn (proposition 2.4). The intervals are paired bootstraps over questions (4000 resamples).

model	version	intact	damaged	lost	gained	excess churn
GPT-3.5	March	63.5	62.4	7.1	5.9	0.118
GPT-3.5	June	65.0	60.6	10.6	6.2	0.124
GPT-3.5: change of the damage cost, June minus March: +3.2 points, 95% interval [-1.5, +8.2]						
GPT-4	March	88.2	85.0	6.8	3.5	0.071
GPT-4	June	84.7	79.4	8.5	3.2	0.065
GPT-4: change of the damage cost, June minus March: +2.1 points, 95% interval [-1.5, +5.6]						

5 Society

Table 3: GSS spending items, 1985–2024. Direct influence of the wording: median absolute change of the share “too little”, in points. Contextual: years in which the point estimates are contextual, for the rank-2 test on “too little” against the rest and for the canonical test on all dichotomizations. Posterior: the largest posterior probability of canonical contextuality over the years (Dirichlet posteriors at the effective sample sizes, 1000 draws). Regimes of the year-to-year changes in each wording: bend / tear? / tear / break.

items	direct A	direct B	contextual (rank 2 / canonical)	posterior	wording X	wording Y
welfare-cities	43.9	26.8	0 / 0 / 0	0.06 / 0.00	10/13/0/0	12/11/0/0
welfare-race	44.7	6.5	0 / 2 / 0	0.80 / 0.00	9/14/0/0	8/14/1/0
health-education	1.6	3.6	3 / 10 / 8	0.97 / 0.93	6/17/0/0	9/14/0/0

Wording is a direct influence. Calling welfare “assistance to the poor” raises the share saying “too little” by a median of 44–45 points, the classic GSS wording effect [29], and calling the cities item “assistance to big cities” changes it by 27. Welfare and cities is noncontextual in every year under both tests. Welfare and race is contextual at the point estimates in two years of the canonical test (posterior probability 0.80 at most). Health and education is the one candidate: contextual at the point estimates in 3 of 24 years for the rank-2 test and in 10 of 24 for the canonical one, with posterior probabilities between 0.37 and 0.97 in those years. Only 1988 exceeds 0.95 (0.968), and none comes near the screen $1 - 0.05/72 \approx 0.9993$ for 72 system-years (with 1000 draws, passing it would require every draw to be contextual). This posterior screen is not a corrected test, and it is optimistic, since it ignores the clustered design. The answer scale is ordered, so we repeated the canonical test with only the two order-respecting cuts (“too little” against the rest, “too much” against the rest): the welfare and race flags disappear, since they come from the split “about right” against the two extremes, and health and education keeps 8 of its 10 flags, with posterior probabilities at most 0.93. We conclude that wording acts as a direct influence everywhere, and that the health and education system lies close to the contextuality boundary in many years without crossing it decidedly in any. Whether the two wordings measure the same content at all is debated [21]; the Contextuality-by-Default reading is agnostic about it.

Tears over four decades. Most year-to-year changes bend or possibly tear: the point estimates of two items often sit near a wall, so small changes cross it without the bands deciding. One tear is decided: for welfare and race in wording Y, from 2014 to 2016, the wall $f_{\text{too little}} = g_{\text{about right}} + g_{\text{too much}}$ is crossed, that is $f_1 + g_1$ passes 1. In 2014 “too little” for “assistance to the poor” (62.5%) and for “assistance to blacks” (26.9%) add to 0.89; in 2016 they add to 1.13. After the tear the margins force at least 13% of respondents to say “too little” to both: a combination that could previously be absent no longer can be. The bands are simultaneous within a state but not across the 138 year-to-year steps; with a Bonferroni correction over the steps this tear is only a possible tear. The step from 2018 to 2021, across the survey’s move to the web, is among the three largest changes in three of six series, but it spans three years and is not among the three largest per year in any.

Table 4: GSS panels, welfare item in each wording (“welfare”, “assistance to the poor”). Agreement of the same respondent’s answer in waves 1 and 2, the maximal agreement the margins allow, the excess churn; the gluing gap of the three waves in bits (plug-in) with a bootstrap interval, against the 95th percentile of the gap under the glued (Markov) law of the answers (400 resamples, none of which reaches the observed gap).

panel	wording	n	agree	max	excess	gluing gap [95%]	Markov null 95%
2006–2010	welfare	592	0.623	0.958	0.334	0.066 [0.047, 0.129]	0.027
2006–2010	the poor	605	0.744	0.950	0.207	0.063 [0.047, 0.118]	0.028
2008–2012	welfare	552	0.623	0.953	0.330	0.067 [0.049, 0.126]	0.030
2008–2012	the poor	641	0.702	0.942	0.240	0.069 [0.051, 0.124]	0.024
2010–2014	welfare	604	0.621	0.939	0.318	0.082 [0.060, 0.143]	0.026
2010–2014	the poor	622	0.664	0.958	0.294	0.055 [0.043, 0.106]	0.025

Churn and composition. The panels (table 4) show the largest masked change in the paper. Between waves the margins of the welfare item barely move, so a maximal coupling would let 94–96% of respondents keep their answer; in fact 62–74% do. A fifth to a third of the respondents change their answer beyond what the shift of the margins requires. The answers of the three waves are not glued: the gap of 0.055–0.082 bits lies above every resample of the Markov law of the answers.

Table 5: Latent-class models of the panels (three classes): likelihood-ratio G^2 against the saturated table with its chi-square p-value, and the gluing gap of the answers that the fitted model implies. Stable: 18 degrees of freedom; drifting and latent Markov: 6. Parametric-bootstrap p-values (100 draws) give the same verdicts.

			observed	stable		drifting		latent Markov	
panel	wording	n	gap	G^2 (p)	gap	G^2 (p)	gap	G^2 (p)	gap
2006–2010	welfare	592	0.066	32.3 (0.020)	0.104	5.8 (0.442)	0.059	5.9 (0.439)	0.058
2006–2010	the poor	605	0.063	77.7 (< 0.001)	0.081	6.6 (0.364)	0.055	9.5 (0.146)	0.041
2008–2012	welfare	552	0.067	36.2 (0.007)	0.105	6.7 (0.354)	0.060	7.8 (0.256)	0.052
2008–2012	the poor	641	0.069	35.8 (0.008)	0.082	5.5 (0.480)	0.065	7.1 (0.313)	0.053
2010–2014	welfare	604	0.082	25.9 (0.103)	0.094	2.9 (0.822)	0.079	8.8 (0.188)	0.053
2010–2014	the poor	622	0.055	18.0 (0.457)	0.067	2.8 (0.836)	0.053	4.5 (0.616)	0.042

Change or unreliability? Unreliable answers around a fixed attitude also produce such a gap: the fitted stable models imply gaps of 0.067–0.105 bits (table 5). The stable model is rejected in the 2006 and 2008 panels for both wordings ($p \leq 0.02$) and not in the 2010 panel ($p = 0.10$ and 0.46); but it forces the answer margins to be equal at all waves, and they drift. Both twenty-parameter models fit all six series ($p \geq 0.15$): a fixed latent partition whose answers drift, and a latent Markov chain answered with one response law. The panels therefore do not show that attitudes fail to compose. They are consistent with attitudes that change as a Markov chain, which glues, seen through answers that are unreliable, which do not. The gluing gap of the answers measures the unreliability as much as the change; three waves of about 600 respondents cannot separate the drifting from the latent Markov reading.

Outcome. The 2008 vote is associated with the first-wave answer, in both wordings: among those saying “too little” for welfare 80.6–81.4% voted for Obama, against 38.1–47.4% of those saying “too much”; for “assistance to the poor” the shares are 68.6–75.0% and 21.3–31.9% (unweighted, panels of 2006, 2008, 2010). In the 2010 panel the vote preceded the first wave, so this is an association, not a prediction.

6 Site separation on the General Social Survey

ThreeLayers [33, §2] assigns to each layer of the programme its own invariant, and its stratification remark states that each is nontrivial on its own site and zero on the others (for Layers 2 and 3 in the precise form of [33, Thms. 15.1, 15.3]): Layer 1 (operations) carries a holonomy class $[g] \in H^1(\Gamma, A)$ of a connection on a graph; Layer 2 (probability) lives on a site with a global joint, where the invariant is information, $I = \delta_t H$; Layer 3 (outcomes) lives on a cover with no global context, where the invariant is a contextuality class and its contextual fraction. The theorems concern sites and coefficients and predict nothing about data; what the survey can show is that the three invariants, computed on the same respondents, are nonzero or zero independently of one another, which is what the stratification requires of them. We use all eleven national-spending items that the GSS asks in both randomized wordings (welfare / the poor, big cities, Blacks, crime / law enforcement, drug addiction / rehabilitation, education, health, environment, space, defense, foreign aid), every year 1985–2024: 55 item pairs in 24 years, 1320 system-years. Between 54.6% and 73.5% say “too little” for “assistance to the poor” and between 13.4% and 33.4% for “welfare” (weighted, by year); 55,169 respondents answered one of the two welfare wordings (27,047 and 28,122), none both.

6.1 Layer 3: contextuality is absorbed by direct influence

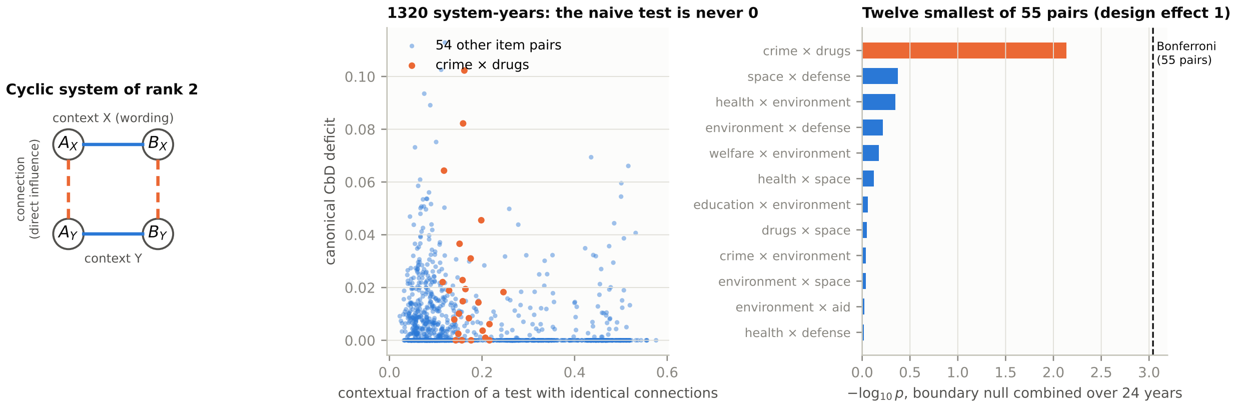


Figure 1: Left: the system of two items in two wordings as a cyclic system of rank 2; solid edges are contexts, dashed edges are connections. Middle: for every system-year, the contextual fraction a test with identical connections would report, against the canonical CbD deficit. Right: the twelve item pairs with the smallest combined p-values for canonical contextuality (boundary null, Fisher over the 24 years, design effect 1); the dashed line is the Bonferroni threshold over 55 pairs.

A test that treats the two wordings as the same measurement, as a Bell-type or Kochen–Specker test does, asks for one law q of (A, B) under both tables. The noncontextual-fraction linear program of [1], applied formally to this model (which, having two different laws on one context, is outside the no-signalling setting for which that fraction is defined), gives $1 - \max\{\lambda : \lambda q \leq T_X,\ \lambda q \leq T_Y\} = \mathrm{TV}(T_X, T_Y)$, positive whenever the wording changes anything. It is positive in all 1320 system-years, from 0.022 to 0.576 (median 0.119; fig. 1). Contextuality-by-Default instead couples each item with itself across wordings maximally and asks whether the rest fits. In canonical form the point estimates are contextual in 523 system-years (307 with only the order-respecting cuts, 93 in the rank-2 reduction): the canonical test is far more sensitive than the binary one. To decide which flags are sampling noise we test each against a noncontextual truth on the boundary: the Y table is moved toward the X table until the deficit vanishes, both tables are simulated there at the effective sample sizes, and the p-value is the share of simulated deficits at least the observed one (200 simulations per system-year; p-values of the form $(k + 1)/201$ and $p = 1$ for a zero deficit, so Fisher's combination below is conservative). Combining the 24 independent yearly samples of each pair by Fisher's method, the smallest p-value is 0.0073 (0.0041 with 500 simulations), for crime and drugs, which is flagged at the point estimates in 20 of 24 years; the next is 0.42. With design effect 2 it is 0.15. No pair passes the Bonferroni threshold 0.05/55. (The Dirichlet posterior used in section 5 is more liberal here: it puts 25 system-years above 0.95, because the posterior of a statistic that is zero on a whole region piles up on the contextual side near the boundary.)

So the expectation that contextuality vanishes once direct influence is accounted for is borne out on the whole survey: the naive test reports contextuality everywhere, and none is detected once direct influence is allowed. Non-detection is not proof of absence: the boundary null is one natural noncontextual truth, not a proved least favorable one, and 523 point estimates are contextual. Crime and drugs is the one system that sits persistently near the boundary: in wording X ("halting the rising crime rate", "dealing with drug addiction") the two items are more strongly associated than the shift of their margins alone allows when they are reworded as "law enforcement" and "drug rehabilitation". This is a candidate for a designed test, not a finding.

6.2 A positive control, and the power of the test

A negative result means little unless the test is shown to detect contextuality where it is known to be present. Cervantes and Dzhafarov's Snow Queen experiment [14] is the standard behavioural case. It is a cyclic system of rank 4 in the Bell design: two choices of character, each paired with two choices of characteristic, 1,799 respondents, one context each. Its direct influences are large, yet the authors find contextuality beyond them. We recomputed it from the published proportions and sample sizes with the programme's code, both the closed-form criterion and the linear program of [8, §4] (table 6). The statistic for correct responses, $s_{\mathrm{odd}} - (n - 2) - \Delta = 0.452$ with 99.99% bootstrap interval $[0.226, 0.668]$, is the published one exactly. For all responses we obtain 0.276, interval $[0.012, 0.494]$, against the published 0.279 and $[0.008, 0.506]$; the difference is probably due to the three-decimal rounding of the published proportions and to Monte Carlo error in the bootstrap. Both systems are contextual under both methods.

This validates the rank-n cyclic criterion and its linear program. It does not by itself validate the pipeline behind the non-detections of sections 7 and 8: the canonical split over all dichotomizations of 3×3 contents with the boundary null. We therefore planted contextuality in two-context 3×3 systems: a depression–anxiety-like joint in one context, and in the other the same joint with its coupling weakened by a margin-preserving perturbation, with or without a shift of the margins (direct influence with a naive fraction of about 0.18, as in the panels). We used 40 replicates of each, 100 null draws, and $500 + 500$ or $1{,}200 + 400$ observations (table 7). The test holds its size:

Table 6: Positive control: the Snow Queen experiment [14], recomputed from the published proportions. The four contexts are taken in cyclic order, two of them transposed so that consecutive contexts share a content. s_{odd}, Δ and the statistic as in Contextuality-by-Default for cyclic systems of rank 4; contextual when the statistic is positive.

responses	s_{odd}	Δ	statistic	99.99% interval	published	verdict (criterion, LP)
correct only	4.000	1.548	0.452	$[0.226, 0.668]$	0.452, $[0.226, 0.668]$	contextual, contextual
all	3.346	1.070	0.276	$[0.012, 0.494]$	0.279, $[0.008, 0.506]$	contextual, contextual

it rejects in 0–7.5% of compatible systems, and in none when the margins differ. Its power is low near the boundary. A population deficit of 0.04 or less is detected at most one time in ten, and only deficits of 0.08 to 0.16 are detected about half the time. The non-detections of this paper therefore rule out large contextuality, not small.

Contextuality is also the hardest layer to activate in observational data, for a reason separate from power. Its premise is that the same contents are measured in different contexts, with the contexts the only difference. The GSS split ballots meet this, since wording is randomized. Sites, ICU types and recruitment waves do not, because their populations differ, and that difference enters as direct influence, which Contextuality-by-Default absorbs by design. So outside randomized designs a non-detection is expected whatever the truth (section 15).

Table 7: Planted check of canonical CbD with the boundary null (`ext/snowqueen/cbd_split_power.py`): rejection rate at 5% in 40 replicates, by population deficit of the planted system. "Shifted": the margins also differ between contexts (direct influence).

	population deficit (margins equal / shifted)				
sizes, margins	0 / 0	0.02 / 0.006	0.04 / 0.016	0.08 / 0.036	0.16 / 0.076
$500 + 500$, equal	0.075	0.05	0.0	0.325	0.55
$500 + 500$, shifted	0.0	0.0	0.0	0.05	0.40
1,200 + 400, equal	0.05	0.025	0.10	0.15	0.575
1,200 + 400, shifted	0.0	0.0	0.0	0.075	0.525

6.3 Layer 1: the wording connection is not flat

In ThreeLayers' bias site [33, §36], a connection on a graph is flat when it is a coboundary, a sum of a lean of each source and an effect of each topic, and its holonomy around a cycle is the loop sum Φ. Here the sources are the two wordings and the topics the eleven items: the graph $K_{2,11}$ has $\beta_1 = 22 - 13 + 1 = 10$ independent cycles, the loops through two items i, j, with $\Phi(i, j) = d_i - d_j$. The connection is flat exactly when the wording shifts the log-odds of "too little" by the same amount for every item, and its class lives in $H^1(K_{2,11}; \mathbb{R}) \cong \mathbb{R}^{10}$ [33, Thm. 13.3]. In every one of the 24 years flatness is rejected (Wald test with a bootstrap covariance, $\chi^2_{10} \geq 333$; Bonferroni over the years). The class is dominated by two items with opposite signs (fig. 2): the log-odds of "too little" are 1.96 lower for "welfare" than for "assistance to the poor", and 1.29 higher for "solving the problems of the big cities" than for "assistance to big cities"; crime (0.51) and race (0.34) follow, and space, defense, environment and health are within 0.07 of zero. Raw d_i are quoted; only their differences, the loop sums, are invariant, and fig. 2 plots $d_i - \bar{d}$. The yearly values are pooled by inverse variance.

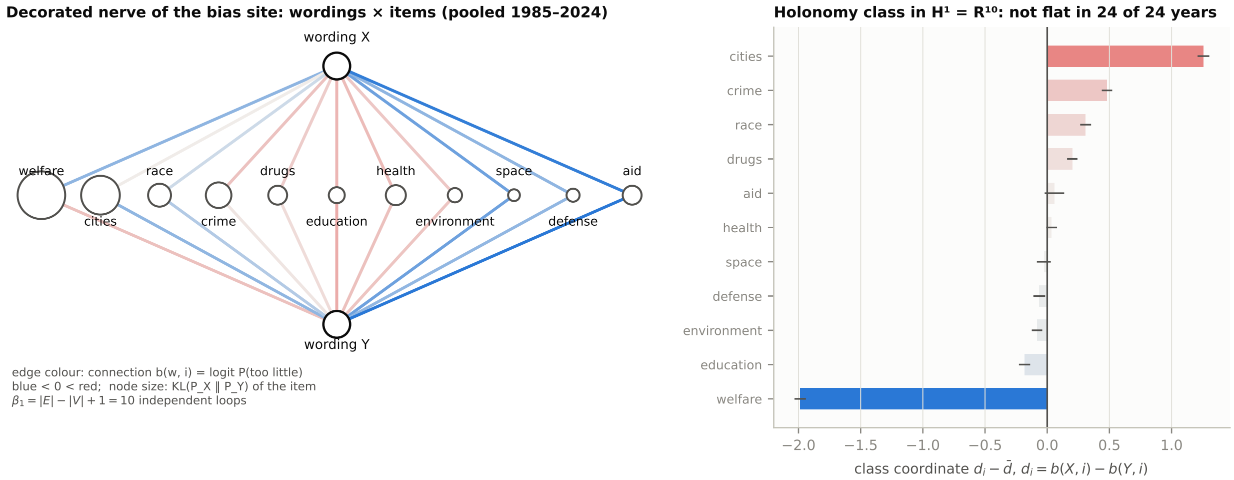


Figure 2: Left: the bias site of the wordings as a graph $K_{2,11}$, wordings joined to items; each edge carries the connection $b(w, i) = \operatorname{logit} P(\text{too little} \mid w, i)$, the mean over the years, and each item is decorated with the Bregman divergence $D_{KL}(P_X \| P_Y)$ of its two wordings. Right: the holonomy class, the wording effects $d_i = b(X, i) - b(Y, i)$ modulo a common constant, pooled over the years with 95% intervals.

6.4 Layer 2: dependence and the entropy brane

Within one wording the two items have a joint law, and its distance from the maximum-entropy coupling with the same margins, the point where the entropy brane meets the constraint brane [8, Thm. 3.1], is the mutual information $I = D_{KL}(T \| f \otimes g)$, the Layer 2 invariant $\delta_t H$. Over the 2640 tables its median is 0.024 bits (largest 0.23), and independence is rejected at a Bonferroni level in 1140 (section 9 shows the brane picture).

6.5 The three invariants dissociate

On the same 1320 system-years no contextuality is detected (Layer 3). The loop through the two items is non-flat in 493 (Layer 1: $|z| > 4.12$, Bonferroni over the system-years, with the variance of $d_i - d_j$ from the bootstrap including the covariance of answers given by the same respondents), the items are dependent in at least one wording in 733 (Layer 2: G-test at 0.05/2640), and both hold in 340, of which 261 are not even contextual at the point estimate. Across system-years the naive contextual fraction tracks the size of the loop (Spearman 0.85) but hardly the dependence (0.24 with the mutual information under X), and the loop and the dependence are almost unrelated (0.14).

Two observations follow, and both are consistent with the stratification rather than tests of it. First, the three invariants are nonzero or zero independently: Layer 3 is not detected where Layers 1 and 2 are nonzero, and neither of these two determines the other. Part of this is structural, since the loop sums are functions of the margins and canonical Contextuality-by-Default absorbs every marginal shift as direct influence; the empirical content is that what remains after the margins, the joint laws within the wordings, does not produce contextuality, while it does carry dependence. Second, the naive contextuality of the wording experiments tracks, to a first approximation, the Layer 1 holonomy of the wording connection, which a test with identical connections misreads as contextuality and which Contextuality-by-Default places outside Layer 3.

6.6 Composition and paths in the panels

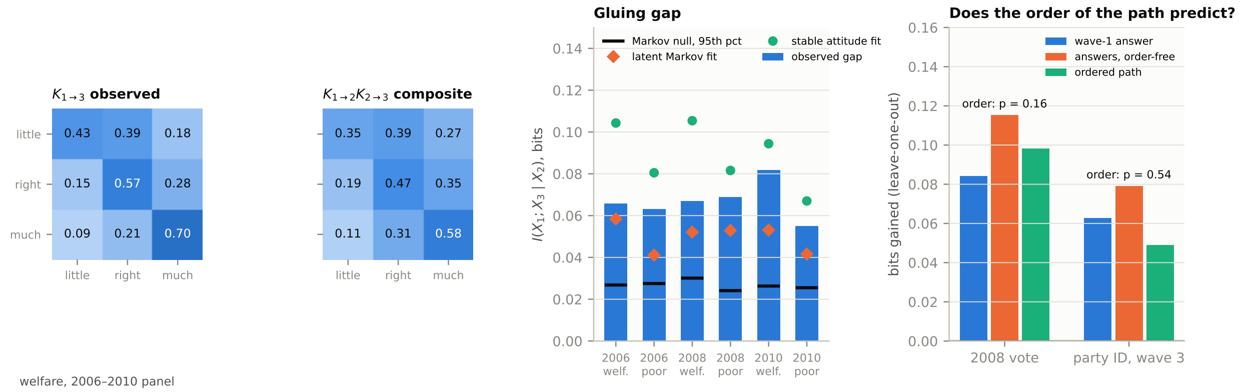


Figure 3: Left: the wave-1 to wave-3 kernel of the welfare answer and the composite of the two wave-to-wave kernels (2006–2010 panel). Middle: the gluing gap of the answers in each series, against the 95th percentile of the Markov null and the gaps implied by the fitted stable-attitude and latent Markov models (table 5). Right: bits per respondent gained over the panel-and-wording baseline in the association with the 2008 vote and wave-3 party identification, of the wave-1 answer, by the three answers without their order, and by the ordered path (leave-one-out); p: permutation test of the order.

The wave-1 to wave-3 kernel is not the composite of the wave-to-wave kernels: their rows differ by 0.04 to 0.19 in total variation, the composite being more diffuse, as happens when the answers carry memory beyond the last wave (fig. 3). This is the gluing failure of table 4 seen in one transition, and, as shown there, a latent Markov chain answered with error reproduces it.

Does the path matter for outcomes? The vote is recalled at or after waves that belong to the path, and party identification is measured at wave 3, so these are associations, not predictions. We compare three summaries of the answers in their association with the 2008 vote (Obama or McCain) and of wave-3 party identification (Democrat, Independent, Republican, leaners with their party): the wave-1 answer, the multiset of the three answers (which answers were given, regardless of order), and the ordered path, by permutation tests that permute outcomes within the cells of the smaller summary (for the multiset against the wave-1 answer, which are not nested, the statistic is a difference of fits, not a likelihood ratio). Pooled over the six series with exact permutation tests stratified by series, the multiset adds to the wave-1 answer ($p < 0.001$ for both outcomes), but the order adds nothing to the multiset ($p = 0.16$ for the vote, 0.54 for party), and out of sample the ordered path fits worse than the multiset (0.098 against 0.115 bits gained for the vote, 0.049 against 0.079 for party). Two of the twelve series-level tests of order fall below 0.05, as expected by chance. Repeated measurement helps because it averages unreliability; the order of the answers carries no further information about these outcomes. This answers the question of path dependence for the survey: $K_{1\to3} \neq K_{1\to2}K_{2\to3}$ for the answers, but the difference does not reach the outcomes.

7 Clinical data

Pre-registration. The analyses of this section after NHANES, and those of section 8, follow written plans dated before any data values were examined: a plan for MIMIC-IV (2026-10-09) and amendments for the open clinical data and for the C19PRC panels (2026-10-10). Each plan fixes the states, thresholds, contexts, outcomes and decision rules. Each analysis program was first run on synthetic data in the layout of its dataset, with planted answers, and recovered them all. Results were frozen with SHA-256 hashes before any replication sample was opened. Every departure from the plans is listed in section 7.6. The analyses of the earlier sections were not pre-registered.

The five analyses. The same five analyses run on every clinical dataset.

A. *Geometry*: regimes between contexts and the toric type of the coupling polygon.
B. *Composition*: the gluing gap of three time points against a Markov null, and the stable, drifting and latent Markov models of table 5.
C. *Path and outcome*: whether the multiset of three states adds to the first, and whether their order adds to the multiset.
D. *Contextuality across sites or units within severity strata*: canonical CbD with the boundary null, combined by Fisher's method.
E. *Selection*: the contextual fraction of three pairwise tables (X, Y), (X, Z), (Y, Z), where Z is measured only in some patients. It is computed available-case and compared with a sampling null. A fraction above the null means the tables are incompatible beyond sampling: measurement of Z is not at random with respect to (X, Y).

The plans also asked for the fraction after inverse-probability weighting on the (X, Y) cells, and decided "selection explains the incompatibility" when it fell below the null. That rule is uninformative, which we noticed only after the results were frozen. Cell weighting makes the measured patients reproduce the full (X, Y) table, and the weighted measured sample is then a joint law of (X, Y, Z) with all three tables as margins, so the fraction after weighting is 0 by construction. We report it, but only the available-case test is evidence. Weights not built on the (X, Y) cells (a logistic model of measurement, or the survey's own weights) can leave a residual. Their fraction is a weak check that those weights capture the selection. In the programme's terms, the question is whether the three tables are margins of one law: whether a point exists in the fibre of the measurement map over them. The deeper reading, selection as a kernel from the population to what is observed, is not used here, because the selection kernel is unknown. Fitting it is exactly the step that cell weighting does trivially.

NHANES. From 2009–2010 to 2011–2012, diabetes among adults went from 10.2% to 11.1% and the BMI classes barely moved. The change is a bend with stress 0.08; among adults 55+ it is a bend with stress 0.90, close to but below the threshold of proposition 2.2, which therefore certifies the bend. The probability of diabetes given obesity is 18% overall and 31% at 55+ in both cycles, and the system is not contextual in either test. Two cycles of a stable population are a negative control: the monitor reports no tear where none is expected.

Medical questions. On USMLE the update moved GPT-4 from 88.2% to 84.7% and GPT-3.5 from 63.5% to 65.0% (intact prompts); the change is a bend apart from the near-empty wall of remark 2.3. The larger finding is table 2: damaged clinical information changes which questions a model gets right far more than it changes how many.

7.1 Intensive care: the PhysioNet 2012 Challenge

The PhysioNet/Computing in Cardiology Challenge 2012 [28, 18] gives the first 48 hours of 4,000 adult ICU stays in each of two independent sets, with in-hospital death.

- States: the hemodynamic state in three windows (hours 0–6, 21–27, 42–48) from the mean arterial pressure, banded below 65, 65–75 and above 75 mmHg; there is no vasopressor variable, so shock is defined by pressure alone.
- Covariates: the first creatinine and lactate of the first day, banded, and mechanical ventilation.

- Contexts: the ICU type (coronary, cardiac surgery, medical, surgical).
- Design: set A is the analysis sample and set B the replication.

The negative control passes: the system (hemodynamic state, creatinine) moves from set A to set B by a bend with stress 0.83, below one.

Table 8: PhysioNet 2012, analysis and replication sets. Gaps in bits; p-values of the latent-class models from the χ^2 reference; the selection row gives the available-case contextual fraction and the 95th percentile of the sampling null.

decision	set A	set B	replicated
answers not glued (gap vs Markov null)	yes (0.0165 vs 0.0039)	yes (0.0175 vs 0.0042)	yes
latent Markov model not rejected (3 time points)	yes ($p = 0.089$)	no ($p = 0.008$)	no
death: multiset adds to the first state	yes ($p = 0.023$)	yes ($p = 0.016$)	yes
death: order adds to the multiset	no ($p = 0.97$)	no ($p = 0.21$)	yes
medical vs surgical ICU contextual, strata H_1	no ($p = 1.0$)	no ($p = 1.0$)	yes
the same, strata SOFA	no	no	yes
lactate tables incompatible (available case)	yes (0.0453; 0.0365)	yes (0.0391; 0.0364)	yes

Six of the seven decisions replicate (table 8). Lactate is measured in about half of the stays. The available-case tables of (first-window state, creatinine, lactate) are incompatible beyond sampling in both sets, so lactate is not measured at random with respect to the first-window state and creatinine. (Weighting on their cells gives 0, by construction.) ICU types differ by decided tears: coronary against cardiac surgery and cardiac surgery against medical in both sets. Within severity strata, no difference between medical and surgical ICUs is detected as contextual. The hemodynamic states do not compose. Three time points do not decide whether a latent Markov chain explains this: it is not rejected in set A ($p = 0.089$) and is rejected in set B ($p = 0.008$). The order of the three states adds nothing detectable to the association with death.

Interchange recheck of the pressure bands. The gaps in table 8 were computed on banded pressures, which the panels showed can inflate them (section 8.1). We reran the interchange test with the window medians in 5-mmHg bins as the raw states. The bin edges include 65 and 75, so banding is an exact merge of 17 bins into 3; the bins are themselves a merge of a continuous value. The plan is `PREANALYSIS_TOWER_ICU.md`, Part 3. On planted chains with 17 states the test held its size (0/20) and had full power (20/20), while the monitor's banded null rejected all 20 first-order chains.

- The bands are not sufficient ($p < 10^{-18}$ in both sets).
- The corrected null is twice the monitor's: 0.0078 against 0.0039 in set A, 0.0068 against 0.0039 in set B.
- The interchange defect carries 38% and 24% of the banded failure.
- The observed gaps, 0.0165 and 0.0175, still exceed the corrected null ($p = 0.0025$ in both sets).

So the ICU conclusion stands, replicated, with a correctly sized null.

7.2 A three-level tower on PhysioNet 2012, and its Fano tubes

This is the first run of the tower of [34] on real data, planned in `PREANALYSIS_TOWER_ICU.md` before any model was fitted.

- *Level two.* Five logistic models of in-hospital death are fitted on one sample: base rate (M0); first-window pressure (M1); pressure and creatinine (M2); those with ventilation and ICU type (M3); SOFA band (M4).

- *Level three.* Each model is transported with a reliability $\alpha \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, \frac{9}{10}, 1\}$, discounted towards the fit sample's base rate. The tower is the posterior over the 30 pairs (model, reliability).
- *Conditioning.* It conditions on an evidence sample one patient at a time. On the first 200 patients it agrees with the programme's tower code to 2×10^{-14}, and coherence $\mu\Omega = \Pi$ holds throughout.
- *Test.* Its predictions are scored on a held-out test sample against flat Bayesian model averaging, which is the same posterior with $\alpha = 1$ only.

On planted data the adapter recovered a planted reliability of $\frac{1}{2}$ and then beat flat averaging, and it did not beat it when the planted reliability was 1.

Table 9: The tower on PhysioNet 2012. Δ: log loss of flat model averaging minus that of the tower on the test sample, with a paired bootstrap 95% interval; level three "earns its keep" when the interval is above 0. The toric surface is that of the 2×3 system death $\times$ first-window pressure band. The third column is exploratory.

	same population: fit A, evidence and test B	shift: fit surgical, evidence and test medical/coronary	shift, reliability floor = evidence base rate
fit / evidence / test	4,000 / 2,000 / 2,000	1,942 / 1,013 / 1,013	same
level 2 posterior	M3: 1.00	M3: 1.00	M3: 1.00
level 3 posterior	$\alpha = 1$: 0.95; $\alpha = \frac{9}{10}$: 0.05	mass on $\alpha < 1$: 0.56	$\alpha = \frac{1}{2}$: 0.50; $\alpha = \frac{3}{4}$: 0.42
Δ (interval)	-0.00001 (-0.00006, 0.00005)	0.00016 (-0.00087, 0.00123)	0.0091 (0.0025, 0.0161)
level three earns its keep	no (as predicted)	no (predicted yes)	yes
predicted / observed death rate	0.141 / 0.141	0.132 / 0.184	0.148 / 0.184
toric surface: tower's posterior	$\mathbb{F}_1$ with 1.00	$\mathbb{CP}^2$ with 1.00	$\mathbb{F}_1$ 0.53, $\mathbb{CP}^2$ 0.47
toric surface of the observed system	$\mathbb{F}_1$, not decided on its bands	$\mathbb{F}_1$, not decided on its bands	same
tears of the level-one tube	none	none	six wall crossings
level-one obstruction of the observed rate	none	$d_1 = 0.107$, equal to the certificate margin	–

Results (table 9).

- *Same population.* Between two random halves of one population the tower behaves as predicted. Level three puts 95% on full reliability and adds nothing to flat averaging, and the predictions are calibrated.
- *Shift.* Models fitted on surgical units and carried to medical and coronary units are the case where level three should earn its keep. As predicted, it moved more than half its mass below full reliability. Against the prediction, it did not improve the forecasts, and they stayed badly calibrated (0.132 against 0.184).
- *Why the shift arm failed.* The tower's own lifting criterion explains it [34, Thm. 10.5, Thm. 10.8]. Every hypothesis predicts a death rate between 0.096 and 0.132, so the observed 0.184 lies outside the admissible set at level one. The certificate measures the gap, $d_1 = 0.107$. Discounting towards the fit sample's base rate can only pull predictions down towards 0.102, while the shift raised mortality. The reliability level as specified cannot represent this shift.
- *An exploratory repair, not pre-registered.* With the floor set to the evidence sample's base rate, level three does earn its keep ($\Delta = 0.0091$, interval above 0). Its posterior concentrates between $\alpha = \frac{1}{2}$ and $\frac{3}{4}$. Calibration improves to 0.148 but is still short of 0.184.

Fano tubes. Each gradation of the tower predicts the death margin f_1 of the 2×3 system death $\times$ first-window pressure band, with the pressure margin g observed. The pair (f_1, g) fixes the coupling polygon, the moment polygon of a toric Fano surface [8, Prop. 8.1]. Here the relevant wall is $f_1 = g_0$, the death rate against the share in shock.

- On one side of the wall the polygon is a triangle ($\mathbb{CP}^2$); on the other it is a quadrilateral ($\mathbb{F}_1$, the blow-up of $\mathbb{CP}^2$ at a point).
- Stacking the level-one polygon over the evidence count gives a tube (fig. 4, left). The level-two and level-three gradations give one polygon per hypothesis, weighted by its posterior (right). The tower therefore carries a posterior over toric surfaces.
- In the same-population arm the tube stays $\mathbb{F}_1$, and the tower is certain of it. Yet the realized system's chamber is *not* decided on its bands. The tower decides the type of the expected system, not of the sampled one.
- In the pre-registered shift arm the tube never leaves $\mathbb{CP}^2$, while the observed system is $\mathbb{F}_1$. The tower is certain of the wrong surface, which is the toric face of its level-one obstruction.
- With the repaired floor the tube runs along the wall and crosses it six times as evidence accrues. Its final posterior is split between the two surfaces, 0.53 and 0.47. That is the honest state, since the observed chamber is undecided on the bands as well.

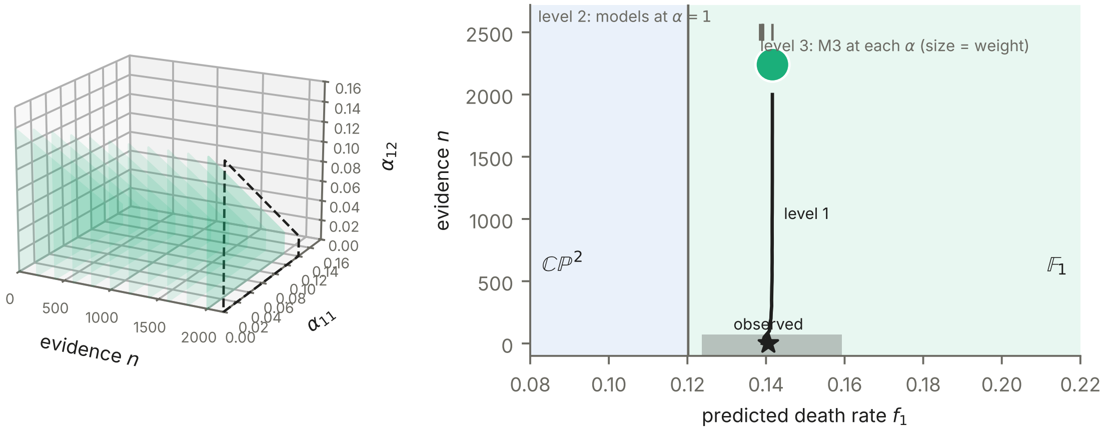
So the toric layer is no longer only descriptive here. It turns a calibration failure into a discrete, certifiable event: the predicted and the observed system lie in different chambers.

7.3 Diabetes: 130 US hospitals, 1999–2008

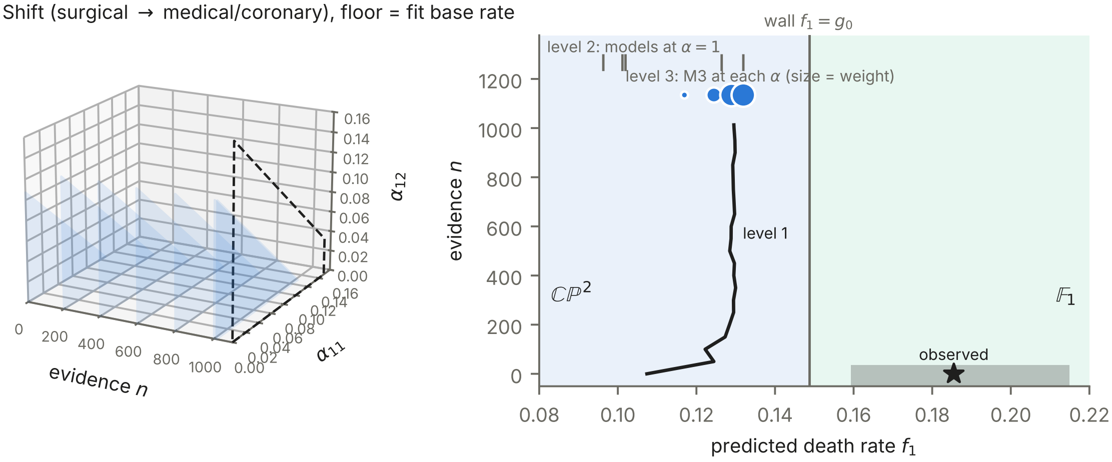
The UCI Diabetes 130-US Hospitals data [32, 35] record 101,766 encounters of 71,518 patients.

- HbA1c was measured in 16.7% of encounters, and the published analysis of these data is about that selective measurement [32].
- With age band and readmission as the other two tables, the available-case contextual fraction is 0.0937 against a null 95th percentile of 0.0135: HbA1c is not measured at random with respect to age and readmission. This is the clearest instance in real data of the selection mechanism of [7, §11]. Cell weighting gives 0 by construction. A logistic model of measurement that also uses admission source and type leaves 0.0081, below the null.
- The data are encounters that met the dataset's inclusion criteria (diabetic inpatient stays of one to fourteen days with laboratory tests and medications), not all admissions of these patients.
- For the 6,339 patients with at least three such encounters, the insulin regimen (none, steady, changed) does not compose across them: gap 0.073 bits against a null of 0.0025. Both twenty-parameter latent models are rejected at 5% ($p = 0.042$ and 0.043), but not at 1%: weak evidence of memory in treatment beyond a latent Markov chain.
- Encounters are ordered by encounter number, assumed chronological. Neither the later regimens (multiset, $p = 0.72$) nor their order adds out of sample to the association with readmission at the third encounter. The order test is nominally significant ($p = 0.035$), but the ordered path has -0.0016 bits of out-of-sample information relative to the base rate (the first state alone gives 0.0002), so by the rule fixed in advance it is not evidence of path dependence.
- Emergency-room and referral admissions differ by a decided tear (insulin use against readmission) and by decided toric surfaces ($\mathrm{Bl}_2\mathbb{CP}^2$ against $\mathbb{CP}^1 \times \mathbb{CP}^1$), but within age bands no contextuality is detected ($p = 1.0$).

Same population (A → B)



Shift (surgical → medical/coronary), floor = fit base rate



Shift, floor = evidence base rate (exploratory)

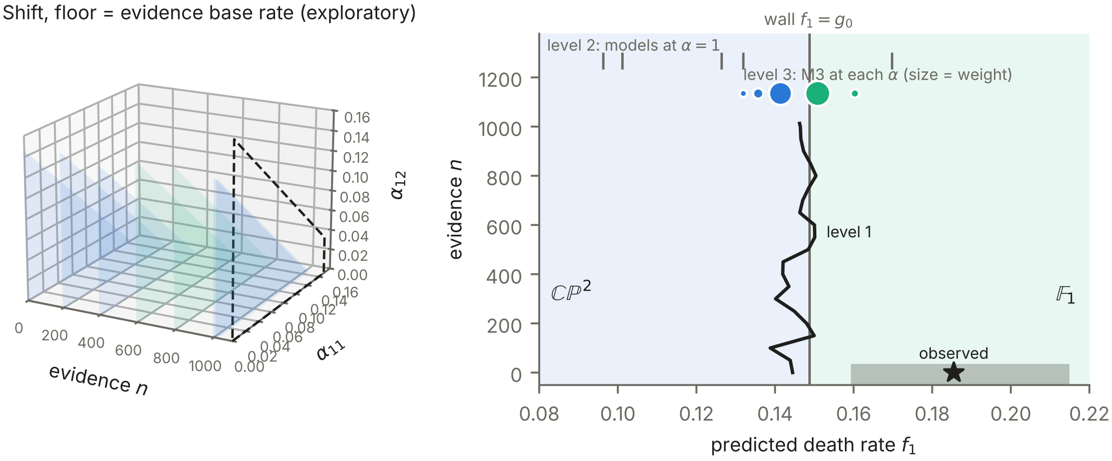


Figure 4: Fano tubes of the PhysioNet tower: same population (top), shift (middle), shift with the exploratory evidence floor (bottom). Left: the coupling polygon of death × first-window pressure band in the coordinates $(\alpha_{11}, \alpha_{12})$ at the tower's level-one prediction, stacked over the evidence count. Colours give the toric surface: blue $\mathbb{CP}^2$ (triangle), green $\mathbb{F}_1$ (quadrilateral). The dashed outline is the polygon at the observed death rate. Right: the same on the axis of the predicted death rate. The level-one path is plotted over the evidence. Level-three hypotheses (M3 at each reliability) are discs sized by posterior weight; level-two models at full reliability are ticks. The star is the observed rate with its band. The wall $f_1 = g_0$ separates the two surfaces.

7.4 Heart disease: four sites

The UCI Heart Disease data [12, 36] come from Cleveland, Budapest, Zurich and Basel, and Long Beach (920 patients), and serve as an illustration.

- Chest-pain type and exercise angina, within diagnosis strata: no contextuality is detected between any pair of sites (smallest $p = 0.11$, Cleveland against Switzerland, where only 8 patients have no disease).
- The sites differ above all in what they measured. Fluoroscopy (`ca`) is missing for 99% of the Hungarian and Long Beach patients and 96% of the Swiss. Thallium scans are missing for 90.5% in Hungary, 83% in Long Beach and 42% in Switzerland.
- The available-case selection fraction is 0.0597. It is descriptive only: no sampling null was computed for these data.

7.5 MIMIC-IV

The plan of [7, §11] for MIMIC-IV [26], made concrete as analyses A–E, is ready to run where the credentialed data are held. It is tested end to end on synthetic data in the MIMIC-IV layout, which it reads directly, and writes only aggregate results with small cells suppressed. It has not yet been run. Its analysis E still reports the cell-weighted fraction, which is uninformative, but it now adds the restricted selection models of section 11.2, each with its own null. Their external covariates are admission type and age band. The plan was amended before any run.

7.6 Deviations from the plans

The second, third and fourth were fixed in the programs before they ran on the real files. The first was found when the UK program first ran. The last two were found after the results were frozen, in an independent review.

- *UK wave membership.* The plan used the archive's `WAVE_1` and `WAVE_2` indicators, but they are empty in the released file. A respondent is counted at waves 1 and 2 if their scores for that wave are present.
- *Irish recontact at wave 5.* Wave 5 has no recontact flag. A wave-5 respondent counts as recontacted if they took part at wave 1.
- *Order of the C19PRC strands.* The Irish plan was written first, as the primary analysis. The UK archive arrived before any Irish value had been examined (only its metadata). The UK amendment then made the UK strand primary and the Irish strand the replication.
- *Heart Disease.* The number of vessels (`ca`) is clipped to 0, 1, 2+, and age is banded < 50, 50–59, 60+; neither is specified in the plan.
- *Four-wave bootstrap.* The frozen four-wave test fitted the observed table with 20 EM starts and each bootstrap table with 4. Fewer starts fit worse, inflating bootstrap G^2 and biasing the p-value upward. We reran with 8 starts for both (section 8); both values are reported.
- *Selection after weighting* is 0 by construction, as explained above; the planted synthetic test could not have shown this, because it planted selection on the weighting cells.

8 Six waves: composition on latent states

Three time points cannot decide whether answers that do not compose are a latent Markov chain seen with error: the latent Markov model is not rejected in some three-wave samples and is rejected in others (tables 5 and 8). With four waves it has 54 nominal degrees of freedom. The COVID-19 Psychological Research Consortium panels [25] supply them. The UK strand (primary) has six waves, March 2020 to Autumn 2021, and 2,025 respondents at wave 1. The Irish strand [31] (replication) has five waves and 1,041 respondents at wave 1.

- State: depression from the PHQ-9 [23], banded 0–4, 5–9 and 10 or more.
- Model: the latent Markov model over four waves, with three classes, time-varying transitions and one response law. It is tested by G^2 with a parametric-bootstrap p-value (200 draws; 100 for the free response law), because the 81-cell table is sparse and some parameters sit on the boundary.
- Check on planted data ($n = 800$): it does not reject a planted first-order chain ($p = 0.44$) and rejects planted second-order memory ($p = 0.033$).
- Power at the actual sample sizes, against the same planted memory (8 replicates, 40 bootstrap draws each): 3 of 8 rejections at $n = 771$ and 0 of 8 at $n = 327$.

Table 10: C19PRC panels, depression (PHQ-9 bands). B1: waves 1–3, latent-class p-values from the χ^2 reference; B2: waves 1–4, parametric-bootstrap p-values; B3: all waves, conditional gaps $I(D_{t-2}; D_t \mid D_{t-1})$ in bits; E: available-case contextual fraction against the 95th percentile of the sampling null.

decision	UK (primary)	Ireland (replication)	replicated
B1 answers not glued	yes ($n = 950$; 0.057 vs 0.016)	yes ($n = 418$; 0.062 vs 0.037)	yes
B1 latent Markov not rejected (3 waves)	yes ($p = 0.50$)	no ($p = 0.042$)	no
B2 latent Markov not rejected (4 waves)	**yes** ($n = 771$; $G^2 = 50.6$, 54 df, $p = 0.89$)	**yes** ($n = 327$; $G^2 = 64.2$, $p = 0.45$)	**yes**
B2, free response law per wave	not rejected ($p = 0.72$)	not rejected ($p = 0.32$)	yes
B2 power against planted memory	3/8	0/8	–
B3 gaps of the answers	0.057–0.099, all above null ($n = 582$)	0.066–0.148, all above null ($n = 271$)	yes
C order adds to the multiset (outcome)	no ($p = 0.57$)	not estimable	–
D recontact vs fresh contextual	not detected ($p = 1.0$)	not detected ($p = 0.48$)	yes
E attrition: tables incompatible	yes (0.0546; 0.0473)	no (0.031; 0.069)	no

In both countries the depression answers fail to compose at every step (table 10). In the UK the gap is larger at later steps, which is descriptive only, since the six-wave group is 582 selected respondents. Yet a latent first-order Markov chain, observed through an error law that is the same at every wave, is not rejected over four waves in either country. With the same number of EM starts (8) for the observed table and for every bootstrap refit, the p-values are unchanged (0.886 and 0.448; free response law 0.723 and 0.317). This is consistent with the reading of section 6: composition fails in the measurement, and the latent states may still compose. It is not proof of it. Against a planted second-order memory of moderate strength, the test rejects in only 3 of 8 replicates at the UK sample size and in none at the Irish one, so longer memory of that size could have been missed, in Ireland almost certainly.

In the UK the three states together are associated with a later report of having *ever* had thoughts of self-harm ($p < 0.001$), but their order is not. The association is partly built in, since item 9 of the PHQ-9 asks about such thoughts. No contextuality is detected between fresh and recontacted respondents ($p = 1.0$ and 0.48); their differences are direct influence, from panel conditioning, attrition or recruitment. Given table 7, only large contextual effects are excluded; the Irish wave-5 deficit of 0.040 ($p = 0.062$) is below what the test can detect. Attrition between waves 1 and 3 makes the UK available-case tables incompatible beyond sampling (0.0546 against 0.0473), so dropout is not at random with respect to wave-1 depression and anxiety. The survey's own wave-3 weights, which are not built on those cells, leave 0.0125. In the smaller Irish strand no incompatibility is detected. Waves 7–9 of the UK strand exist only as a report and are not used.

8.1 Does banding create the failure? The interchange law on raw scores

Every composition test above was run on banded states. Banding is not neutral: it is a merge, a deterministic non-injective kernel, and the interchange law [3, Prop. 6.1] says when gluing commutes with such a kernel. Write α and β for the raw (D_1, D_2) and (D_2, D_3) couplings of the PHQ-9 totals (0–27), and b for the merge onto the three bands. Then

- $L = b^{\mathsf{T}}(\alpha \cdot \beta)b$ is the banded image of the raw Markov composite;
- $R = (b^{\mathsf{T}}\alpha b) \cdot (b^{\mathsf{T}}\beta b)$ is the banded Markov composite, the monitor's null;
- for the observed banded (D_1, D_3) table O, the identity $O - R = (O - L) + (L - R)$ is exact.

It splits the banded composition failure into the raw failure pushed through the bands, $O - L$, and the interchange defect created by banding alone, $L - R$, which vanishes when the bands are sufficient. A raw first-order Markov chain has $O \approx L$ but $L \neq R$. So the theorem predicts that the monitor's banded null is the wrong null whenever the bands are not sufficient. This is the first place where a theorem of the deeper layers changes a data analysis, and it was suggested by the theory, not by the raw data. The plan (`PREANALYSIS_BANDING_INTERCHANGE.md`) was written after the banded results were known but before any raw score was examined, so the test is a post-hoc amendment. The correct reference is a parametric bootstrap from the raw Markov chain, banded afterwards.

Calibration first. On planted raw chains with 28 states and $n = 950$, in 20 replicates each:

- for a genuine first-order raw chain, the monitor's banded null declares a composition failure in all 20, while the interchange test does so in 1;
- against raw chains that return to the wave-1 score with probability 0.2 or 0.6, the interchange test rejects in 17 and 20.

So on banded data the monitor's composition test, as used in tables 4, 8 and 10, is anti-conservative, and the interchange-corrected test holds its size.

Table 11: The interchange test of banding (waves 1–3). Gaps in bits; nulls are 95th percentiles of 400 draws. The total variation of the banded failure $O - R$ splits exactly into the raw part $O - L$ and the interchange defect $L - R$, with "share" the component of the failure along the defect.

	UK ($n = 950$)	Ireland ($n = 418$)
banded gap $I(B_1; B_3 \mid B_2)$	0.0567	0.0617
monitor's null (banded Markov chain)	0.0164	0.0367
interchange-corrected null (raw Markov chain, banded)	0.0231 (median 0.0133)	0.0554 (median 0.0325)
p against the corrected null	0.0025	0.022
$\frac{1}{2}\|O - R\|_1$, $\frac{1}{2}\|O - L\|_1$, $\frac{1}{2}\|L - R\|_1$	0.082, 0.065, 0.017	0.082, 0.058, 0.030
share of the failure along the interchange defect	0.21	0.34
bands sufficient for D_1 / D_3 (p, homogeneity within bands)	4×10^{-6} / 2×10^{-8}	8×10^{-5} / 9×10^{-5}

Results. The bands are not sufficient in either country (table 11). The interchange defect is a circulation, as the theorem says, and carries a fifth of the banded failure in the UK and a third in Ireland. The correct null is larger than the monitor's, by 41% in the UK and 51% in Ireland.

The observed gap still exceeds it in both countries ($p = 0.0025$ and 0.022), so banding does not explain the whole failure: the raw totals do not compose either. That is what the latent model of B2 predicts, since a raw total is itself a noisy measurement, and an error kernel is stochastic, not a merge, so interchange cannot remove it. The two readings agree. Banding inflates the apparent failure but does not create it, and what remains is consistent with measurement error on a Markov latent state. In Ireland the margin is narrow. The raw-score gaps themselves (1.24 and 1.49 bits) are dominated by the sparsity of $28 \times 28 \times 28$ tables and are not interpreted.

9 Toric geometry, the entropy brane and Floer theory on public data

Bridges [8] realizes the coupling polytope $\mathrm{Cpl}(f, g)$ of two margins as the moment polytope of a compact toric manifold $X_{f,g}$, the maximum-entropy coupling as the intersection of the entropy brane with the conormal brane of the marginal constraint [8, Thm. 3.1], and the Floer-nontrivial couplings as the critical points of the Cho–Oh disc potential (§8 there). Here every one of these objects is computed for margins taken from the data.

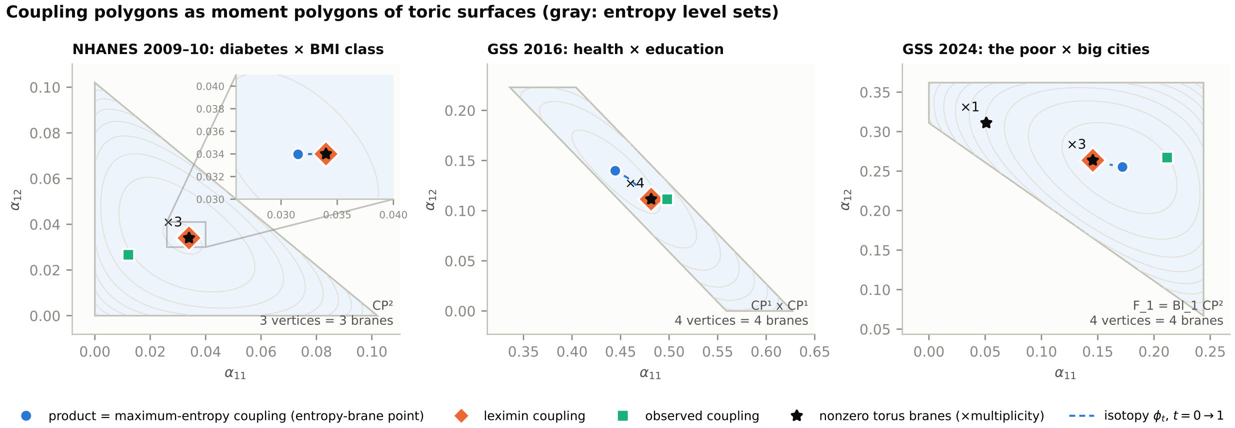


Figure 5: Coupling polygons of 2×3 systems with margins from the data, in the coordinates $(\alpha_{11}, \alpha_{12})$. Each polygon is the moment polygon of the toric surface named in its corner; gray curves are level sets of the entropy, whose maximum is the product coupling. Stars are the valuations of the critical points of the disc potential (nonzero torus branes, with multiplicity), computed exactly from Newton polygons [8, Thm. 8.2]; the dashed curve is the critical point of $\phi_t = (1-t)H + t \log \prod p$ for $t \in [0, 1]$, the isotopy from the entropy brane to the standard brane [8, Prop. 8.7].

Toric surfaces of real margins. For 2×3 margins the toric surface is $\mathbb{CP}^2$, $\mathbb{CP}^1 \times \mathbb{CP}^1$, $\mathbb{F}_1$, $\mathrm{Bl}_2\mathbb{CP}^2$ or $\mathrm{Bl}_3\mathbb{CP}^2$, read off from which cells are facets [8, Prop. 8.1] (fig. 5).

- NHANES, diabetes × BMI class: $\mathbb{CP}^2$ in both cycles, decided on the bands, because only the cells of the rare row (diabetes) can vanish. All three nonzero branes lie over the leximin coupling, which spreads the diabetics equally over the three BMI classes ($\alpha_{1j} = f_1/3$); the product coupling carries none; and the observed coupling lies far from both, with diabetes concentrated among the obese ($\alpha_{11} = 0.012$ against 0.032 under independence in 2009–2010).
- GSS, health (too little or not) × education, wording X: $\mathbb{CP}^1 \times \mathbb{CP}^1$ in 15 of 24 years, $\mathrm{Bl}_2\mathbb{CP}^2$ in 6 and $\mathbb{F}_1$ in 3; the chamber, hence the surface, is decided on the bands only in 2010, 2012,

2014, 2016 and 2021, all $\mathbb{CP}^1 \times \mathbb{CP}^1$. The leximin coupling carries four branes in every year: all of them in the 18 years with four vertices and four of five in the Bl_2 years.
- GSS, “the poor” (too little or not) × “big cities”, wording Y: $\mathbb{F}_1$ in 18 years, Bl_2 in 4, Bl_3 in 2 (1994, 1996), never decided. The leximin coupling carries 3 or 4 branes, at least half of them in every year (for example 3 of 4 in 12 years and 3 of 6 in one).

In all 50 real 2×3 systems the product coupling carries no brane, consistent with [8, Thm. 8.2(iii)], which excludes it whenever the smallest facet product is unique, and the leximin coupling carries at least half of the branes: on public margins, as on synthetic ones, the Floer-theoretic distinction among couplings is robustness against breaking, not entropy. Figure 6 shows the surfaces year by year; most changes of surface are not decided, which is the toric face of the observation that most survey steps are possible tears only.

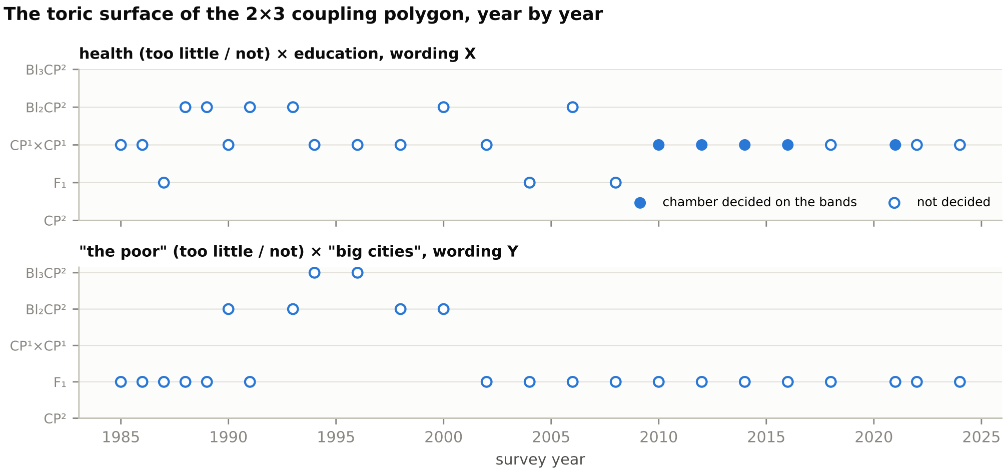


Figure 6: The toric surface of two GSS 2×3 systems, 1985–2024. Filled: the chamber of the margins, which determines the surface, is decided on the 95% bands; open: it is not.

3×3 systems and tears as blow-ups. For 3×3 margins $X_{P,Q}$ is a toric 4-fold, always semi-Fano and Fano only when the polytope has 9 vertices, and its disc potential has no corrections [8, Prop. 8.3, Thm. 8.4]. Over all 2640 GSS tables (pairs, years, wordings) the polytope has 9 vertices in 54, 12 in 1026, 15 in 1070 and 18 in 490. The one decided tear of the survey, welfare × race in wording Y from 2014 to 2016, takes the polytope from 15 to 12 vertices (fig. 7). Exactly one wall changes side, $f_1 = g_2 + g_3$, and the facet $\alpha_{11} = 0$, a tetrahedron with four vertices, disappears: the toric 4-fold is blown down at a torus-fixed point, the 4-dimensional counterpart of the blow-downs of [8, Prop. 8.1]. The branes of 3×3 systems are not computed here.

The entropy brane on data. Restricted to the fibre of a 2×2 system, a segment, the entropy brane is the graph of the derivative of H along the fibre, and the conormal brane is the zero section; they meet once, transversally, at the product coupling (fig. 8, left). The distance of the observed coupling from that point is the mutual information, the Layer 2 invariant (middle). The Hessian of $-H$ on the fibre at the intersection is the Fisher form [8, Prop. 2.1, Thm. 3.1]. Its smallest eigenvalue is positive, which is the transversality; its largest, the steepness of the intersection, diverges as a margin goes to zero, where the brane disappears and the system breaks (for 2×2 systems the fibre is a segment and the two coincide). Over all 2640 GSS 3×3 tables the steepness ranges from 10 to

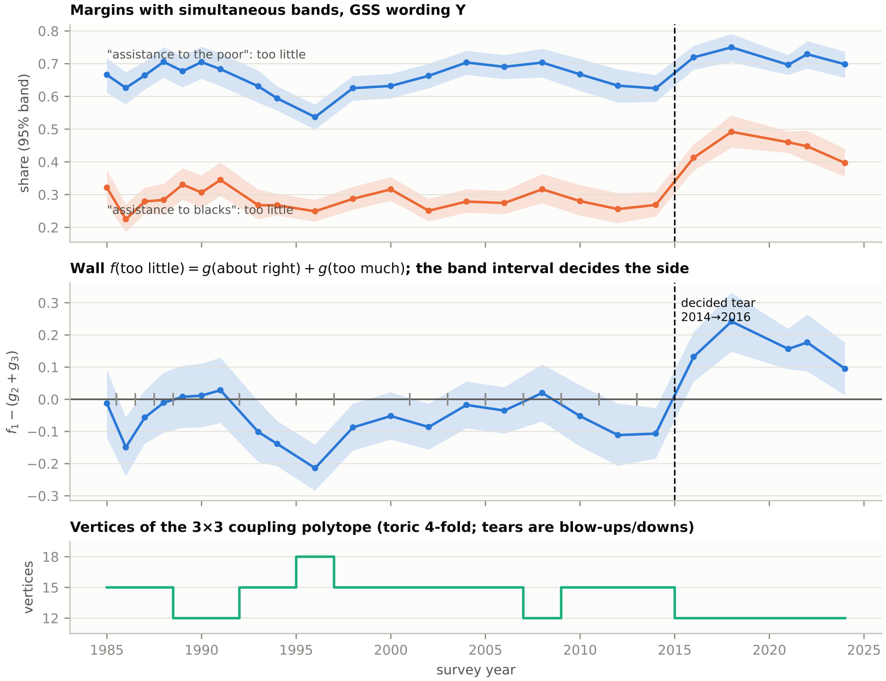


Figure 7: Forty years of one GSS system: “assistance to the poor” and “assistance to blacks”, wording Y. Top: the “too little” margins with their simultaneous 95% bands. Middle: the wall function whose sign changes at the tear, with the interval it takes over the bands; the tear is decided when the intervals on both sides exclude zero; ticks mark possible tears. Bottom: the number of vertices of the 3×3 coupling polytope.

614 (median 50, right panel); it is largest where two rare answers meet, as for education × health in 1990, where few say "too much" to either and the smallest product of margins is 0.0008. These are the systems closest to breaking, although none breaks on its bands.

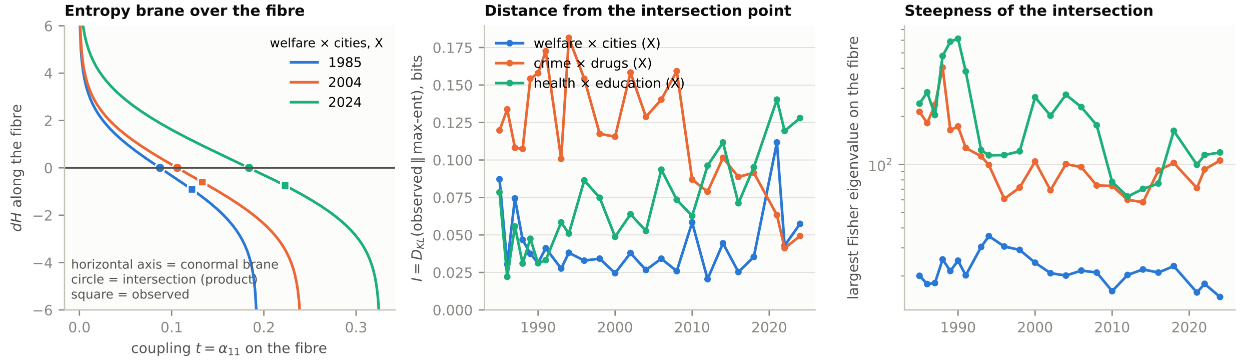


Figure 8: Left: the entropy brane over the fibre of welfare × cities ("too little" or not, wording X) in three years, meeting the conormal brane (the horizontal axis) at the product coupling; squares mark the observed couplings. Middle: mutual information, the divergence of the observed coupling from the intersection point, for three item pairs (3 × 3). Right: the steepness of the intersection, the largest eigenvalue of the Fisher form on the fibre (log scale).

Dual affine geometry. The Fisher metric is the Hessian of $-H$ [8, Prop. 2.1], and its two flat structures, the exponential (e) and mixture (m) coordinates, are Legendre dual [4]. For an m-flat family M and a law q, the e-projection p^* of q onto M satisfies $D(p\|q) = D(p\|p^*) + D(p^*\|q)$ for every $p \in M$ [13]. Figure 9 applies this to the two wordings of the welfare item in 2024: with q the "welfare" law, p the "assistance to the poor" law and M the laws with p's share of "too little", $D(p\|q) = 0.437$ bits splits into 0.413 bits carried by the shift of "too little" and 0.024 bits of reallocation between "about right" and "too much". So 94% of the information in the wording effect is the shift of a single category.

Why these layers are descriptive here. Toric surfaces, branes, the entropy brane and its steepness are computed exactly on real margins. But nothing in these data asks a question whose answer depends on them beyond what the margins already give. A toric type is a function of the chamber, and the chamber of the margins. The steepness grows as a margin approaches zero, which the bands already see. The Floer data concern pairs of couplings, and no analysis here compares couplings across contexts in a way that needs them. To activate these layers one must test a prediction: for example, that systems with steep entropy branes break sooner, or that the toric type at one time predicts the next tear better than the distance to the nearest wall. We have not done so (section 15).

10 The structural backbone on synthetic data

The data of the earlier sections test bands, regimes, contextuality, holonomy, dependence, composition and selection. They cannot test the programme's algebraic backbone: the monad of kernels and its Kleisli composition, the three-level tower, its mixed distributive law, and the interchange

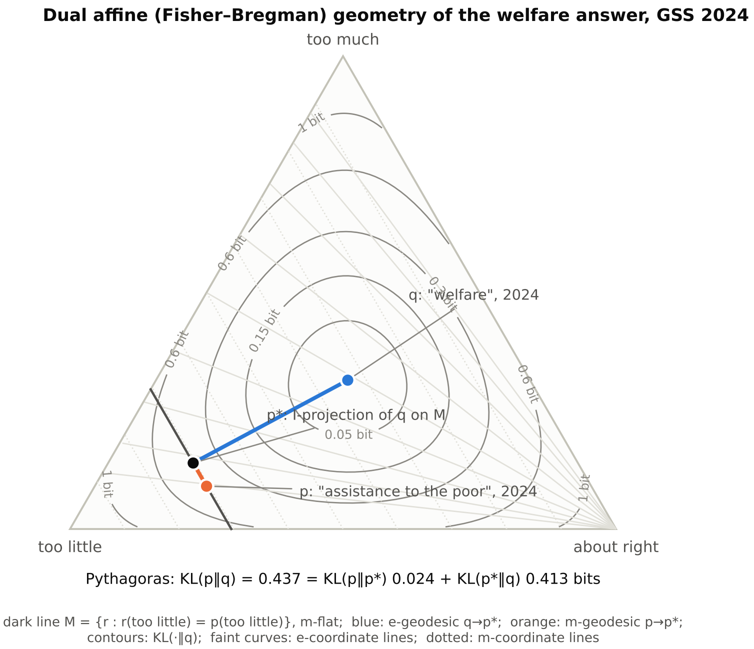


Figure 9: The simplex of answers to the welfare item, with level sets of $D(\cdot\|q)$, the e- and m-coordinate lines, and the generalized Pythagorean decomposition at the e-projection p^*.

of couplings with kernels. No public dataset labels its monads or contains a planted failure. We therefore built four synthetic tests, T1–T4, under three conditions:

- every expected value is a closed form, either stated in a programme paper or given by Bayes' rule in exact rational arithmetic, never a Monte Carlo estimate;
- every planted failure is localized, at a known kernel, axiom, 2-cell or tower level;
- every failure has a magnitude parameter that runs from "the law holds" to "the law fails by a known amount".

The plan, with all expected values, was written before any data were generated. One amendment was made before the first run, because the plan had placed a defect where unitality makes it vanish identically. The data were then generated from a seed and frozen with SHA-256 hashes, and the results frozen after a single run of each program.[1] The tests call the programme's existing code for the tower, the laws and the certificates, and check the new matrix library against the existing whiskering code (agreement to 1.1×10^{-16}). Of the 41 pre-registered expectations, 40 hold (table 12); the exception is a lesson about random-instance checks. These layers are tested on synthetic data because, as section 15 explains, the observational data do not supply the model spaces and channels they act on; the interchange law is the exception, and section 8.1 applies it to real panels.

10.1 T3: Kleisli composition and its known failures

The library holds 15 kernels on three points, each labelled before the run with its class:

- the six permutations;
- two constant kernels;
- three stochastic kernels with full support;

---

[1] The T3 program crashed once, on a vector bug in a helper, before writing any result; it was fixed and run once to completion.

Table 12: Structural tests on synthetic data with closed-form ground truth (`ext/structure/`). "Planted" rows are failures the papers predict; a row passes when the code reports the failure at the stated place with the stated size.

test	claim	expected	observed
T3	Kleisli composites, associativity, unit, $k^{\#}\mu = \mu\,\mathcal{D}(k^{\#})$	exact on the library	195 composites, 2,197 triples, exact
T3	invertible kernels; breaking and healing [7, Props. 5.1, 6.4]	the 6 permutations; zero columns break	as predicted, every kernel
T3	contraction by the Dobrushin coefficient	$\mathrm{TV}(K^{\mathsf{T}}p, K^{\mathsf{T}}q) \leq \eta(K)\,\mathrm{TV}(p,q)$	holds on all 15×190 pairs
T3, planted	whiskering functorial for deterministic kernels only [9, Thm. 10.2]	defect 0 for deterministic pairs	0 on 48 pairs; nonzero on all 49 others
T3, planted	functoriality defect of h_ε at the swap coupling	sympy closed form	$-2\varepsilon^2(1-\varepsilon)^2$, exact at 6 values
T3, planted	multiplicativity defect [7, Prop. 6.6]	$t(\varepsilon)(1-2s)(1-2r)$, $t(\frac{1}{10}) = \frac{369}{2500}$	exact on 216 cases; constant-kernel formula on 380
T3, planted	integrality [8, Thm. 6.1]	integral iff deterministic or constant	the 10 predicted kernels, no others
T1	sequential tower = batch Bayes; coherence at every level	exact	200 of 200 steps
T1	[7, Ex. 9.1] reproduced; entropy split	table to 3 decimals	all 32 entries
T1, planted	Cromwell break at level two; reliability prevents it	weight exactly 0 from the first 2; never 0 with $\alpha < 1$	as predicted; odds bound of [7, Prop. 9.2] holds
T1, planted	claims failing first at levels 1, 2, 3 [34, Thm. 10.5]	levels $1, 2, 3$; $d_2 = \frac{1}{2}$, $d_3 = 2$	levels $1, 2, 3$; $d_1 = 0.4 =$ certificate; $d_2 = \frac{1}{2}$; $d_3 = 2$
T2	[34, Table 2]: shift, averaging and base-relative laws	fail (U, M, Cm), (Cm), none	30/30 violations exactly where predicted
T2, planted	perturbed barycenter β_ε fails (M) only	defect $\frac{\varepsilon}{6}\lVert \frac{\rho_1+\rho_2}{2} - \rho_3 \rVert_1$	exact at 6 values; (U), (Co), (Cm) intact
T2, planted	random instances detect β_ε for every $\varepsilon > 0$	violations iff $\varepsilon > 0$	**none at $\varepsilon = 1$**
T4, planted	interchange [3, Prop. 6.1]	holds for bijections and sufficient merges	holds; fails for the other 3 kernels
T4, planted	sweep $l_t = (1-t)I + tJ/2$	$\frac{1-(1-t)^2}{4}\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$	exact at 6 values

- two with a zero column;
- two merges onto two points;
- a binary family $h_\varepsilon = \bigl(\begin{smallmatrix} 1-\varepsilon & \varepsilon \\ \varepsilon & 1-\varepsilon \end{smallmatrix}\bigr)$.

The monad laws hold exactly. Kleisli composites computed by the distribution-monad code of the tower paper agree with the matrix product on all 195 composable pairs, and Kleisli extensions commute with the barycenter.

The planted failures are all reproduced with their stated sizes:

- *Whiskering in the kernel.* Copy whiskering is functorial on all 48 pairs of deterministic kernels and fails on all 49 pairs of non-deterministic ones, constant kernels included. On the binary family the defect at the swap coupling is $\mp 2\varepsilon^2(1-\varepsilon)^2$ per entry. It vanishes only at the identity and the swap, and is largest, $\frac{1}{8}$, at the constant kernel.
- *Unitality.* Copy whiskering is unital for every kernel, which is why the defect vanishes at the diagonal coupling. The product action is unital exactly for the eight deterministic kernels.
- *Multiplicativity.* The defect formula of [7, Prop. 6.6] holds exactly on all 216 grid cases, and its constant-kernel form on all 380 pairs.
- *Integrality.* Integrality holds for exactly the ten deterministic or constant kernels. So constant kernels are on the good side for integrality and on the bad side for whiskering, as [8, Rem. 6.3] says.

10.2 T1: the tower under a known generative model

The generative model is the four models of a three-valued outcome given in the plan. One of them, $\rho_1 = (\frac{1}{2}, \frac{1}{2}, 0)$, has a zero. Each model is discounted towards the uniform law with reliability $\alpha \in \{1, \frac{9}{10}, \frac{7}{10}\}$, and 200 observations are drawn from ρ_3 at $\alpha = \frac{9}{10}$. The hypotheses are the twelve pairs (model, reliability).

- Level two is the posterior over models.
- Level three is the posterior over reliabilities, each carrying its own level-two state, $\Omega = \sum_\alpha \pi(\alpha \mid x)\, \delta_{\Pi_\alpha}$.
- Level one is the predictive law.

Coherence. Conditioning one observation at a time with the tower code agrees exactly with the batch closed form at all 200 steps, at every level. Coherence $\mu\Omega = \Pi$, $\mu\Pi = \rho$ holds throughout. Conditioning commutes with pooling [34, Prop. 9.12]. After 200 observations the true model has posterior 1.000, while the reliabilities 1 and $\frac{9}{10}$ are not yet separated (0.34 and 0.49).

The Cromwell break. The table of [7, Ex. 9.1] is reproduced in all 32 entries. On the frozen data the first observation is a 2. In the tower without reliability, the zero model's weight is exactly 0 from that step on and never returns: a zero at level one becomes a break at level two. With reliability below one no weight ever reaches zero (the smallest is 1.1×10^{-35}), and every odds ratio moves by at most the bound of [7, Prop. 9.2].

Planted claims at each level. Three claims are each detected first at the planted level, with the distance of [34, Thm. 10.8, Prop. 10.9]:

1. The removed model, asserted after the break, fails at level one. Its distance 0.4 equals the margin of the optimal certificate.
2. The pre-break level-two state, here the prior, has a barycenter $(\frac{3}{8}, \frac{2}{5}, \frac{9}{40})$ inside the hull of the surviving models. So it passes level one and fails first at level two, with $d_2 = \frac{1}{2}$, twice the weight of the removed model.

3. Collapsing the reliability level into one second-order state, δ_Π, passes levels one and two and fails first at level three, with $d_3 = 2$.

As controls, the true tower lifts coherently, and the discounted zero model is admissible.

10.3 T2: the mixed distributive law

The paper's table. The three laws of [34, Table 2] were run on 30 random instances each, generated by the tower paper's own scripts. Violations occur exactly where the table says, 30 of 30 each time:

- the shift law fails (U), (M) and (Cm);
- the averaging law fails only (Cm);
- the base-relative law fails none.

On the paper's counterexamples each failure has size 2 in ℓ^1 at level one, and the shift law's failures vanish after the barycenter, as [34, Thm. 8.3] states.

A planted law. We replaced the barycenter of the context algebra by $\beta_\varepsilon = (1-\varepsilon)\beta + \varepsilon \cdot$ (uniform average over the support). This keeps the unit law, naturality and coherence, so the perturbed law should fail only the multiplication axiom (M). It does. On the pre-registered instance the level-one defect is $\frac{\varepsilon}{6}\|\frac{\rho_1+\rho_2}{2} - \rho_3\|_1 = \varepsilon/126$ exactly, for every ε.

The failed expectation. At $\varepsilon = 1$, the random-instance check finds no violation in 20 instances, although the law fails (M). The generator of the existing checks draws two-point samples, and a uniform average over supports of equal size is associative. So that generator has no power against this failure. A random-instance check that passes is evidence only against the failures its generator can produce (section 17).

10.4 T4: interchange of couplings and kernels

The identity $k^\top(\alpha \cdot \beta)m = (k^\top \alpha l) \cdot (l^\top \beta m)$ of [3, Prop. 6.1] behaves as stated:

- It holds exactly for bijections, and for a merge that is sufficient for the first coordinate. This was checked both with identity outer kernels and with stochastic ones.
- It fails for the same merge on generic couplings (ℓ^1 defect 0.18), for a constant kernel (0.18) and for a stochastic one (0.18). Stochastic outer kernels contract the defect to 0.004–0.006.
- The defect is a circulation every time.
- Along $l_t = (1-t)I + \frac{t}{2}J$ the defect is $\frac{1-(1-t)^2}{4}\left(\begin{smallmatrix}-1 & 1\\ 1 & -1\end{smallmatrix}\right)$ exactly, 0 only at the identity.

Vertical composition is coherent: gluing is strictly associative and unital on 216 triples, and cost is subadditive on 100 pairs.

What these tests show. They show that the code implements the structures the papers state, and that each failure the papers predict appears where predicted, with the predicted size. They do not prove the theorems beyond the finite cases checked; the proofs do that. Nor do they show that any of this structure matters in data; that is the subject of the earlier sections, and of section 17.

11 Forecasting, selection, events and the tower's reliability

The preceding sections left four questions open. They were pre-registered together in one plan (`PREANALYSIS_PHASE4.md`), checked on planted data and run once:

1. Do toric type, chamber transitions or brane steepness forecast anything beyond the margins?
2. Which selection models, not built on the triangle cells, fit the data?
3. Do structural events track external states?
4. Can a richer reliability level repair the tower's failure under a shift?

Two deviations were recorded. A planted design in the selection check was changed before the real run (below). The structural-event analysis was run in the same command as its first planted check; that check's planted confounding was too weak, and a stronger check was then run and passed.

11.1 Does geometry forecast beyond the margins?

Toric type, chamber and steepness are deterministic functions of the margins. So the right baseline is not a linear model in the margins but a flexible one (B2: margins with all squares and products). The question is whether these particular functions add to a generic nonlinear function of the same margins. Geometry (S2 = B2 plus the geometric features) is said to forecast beyond the margins when its test log-loss gain over B2 has a bootstrap interval above 0. The adapter detected a planted geometric signal and correctly found none when the signal was purely marginal.

Clinical deterioration. For each ICU stay and each day, the hourly mean arterial pressure band and heart-rate band form a 3×3 system whose margins are the shares of hours.

- *Features.* Vertex counts and their change (blow-ups and blow-downs), wall flips between the two days, the tear margin, break counts and the steepness of day two.
- *Fit and test.* Fit on set A (3,758 stays), tested on set B (3,802 stays, 549 deaths).
- *Death.* Geometry adds nothing: gain 0.0004, interval $(-0.0020, 0.0027)$; over the linear baseline, 0.0005, $(-0.0014, 0.0026)$.
- *Onset of shock on day two.* Nothing either: 0.0002, $(-0.0006, 0.0009)$.

The reason is visible in the data. 98.8% of day-two systems have a zero margin, typically no hours in one pressure or rate band, and half have a coupling polytope that is a single point. Per-stay systems of a day live almost entirely in the breaking regime, where the toric layer has nothing to describe. A finer time grid or a longer window would be needed to give these systems a nondegenerate geometry.

Societal transitions. The GSS item-pair systems (110 systems in two wordings) were followed from one survey year to the next. The event is a change of the point-estimate chamber. Decided tears are too rare to forecast (one in the paper). Training used steps ending by 2004 (1,320) and testing those from 2006 on (1,100), with intervals clustered by system. Here geometry forecasts beyond the margins: gain 0.041, interval $(0.020, 0.062)$; over the linear baseline 0.038, $(0.017, 0.058)$. Most of it is carried by two quantities:

- the tear margin: the chamber changes at the next survey in 71% of the systems in the quartile nearest a wall, and in 33% of the farthest;
- a change at the previous step: 66% against 42%.

The event is a change of the *point estimate*, and a system near a wall flips by sampling noise as well as by change. So this shows that the programme's distance to the walls predicts chamber instability, which raw margin shifts do not. It does not show that geometry predicts substantive tears.

11.2 Restricted selection models

Weighting on the cells of the fully observed table makes the triangle compatible by construction (section 7). The restricted models each have their own null: tables compatible and selection following the model, simulated and refitted at every draw (table 13). Their weights depend on:

- one content only;
- covariates outside the triangle;
- one content plus those covariates.

One fact limits every such test. Any weights that reproduce the full (X, Y) table make the weighted measured sample a joint law with the three tables as margins, so the tables are compatible. So these tests check only what each model implies about the (X, Y) composition of the measured units. Selection that biases Z within (X, Y) cells, on Z itself or on a covariate that drives Z but not (X, Y), is invisible to the triangle. The planted check had to be redesigned for this reason before the real run. Its first version planted selection on a covariate independent of (X, Y), which leaves the triangle compatible whatever the weights.

Table 13: Restricted selection models: weighted contextual fraction against the 95th percentile of its own null (200 draws); "fits" when at or below. Cell weighting gives 0 by construction and is omitted.

	available case	X only	Y only	external	X + external
PhysioNet A: lactate; X pressure, Y creatinine; ICU type, ventilation	0.045	0.040 / 0.016	0.026 / 0.016	0.068 / 0.021	0.062 / 0.019
PhysioNet B (replication)	0.039	0.033 / 0.015	0.031 / 0.015	0.055 / 0.022	0.049 / 0.020
Diabetes: HbA1c; X age, Y readmission; admission source and type	0.094	0.016 / 0.007	0.093 / 0.008	0.097 / 0.009	0.018 / 0.008
C19PRC UK attrition: D_3; X D_1, Y A_1; age, gender, education, urbanicity	0.055	0.012 / 0.017	0.022 / 0.017	0.020 / 0.023	0.010 / 0.021

- *UK panel.* Dropout between waves 1 and 3 is consistent with selection on wave-1 depression alone, on demographics alone, or on both. It is not consistent with selection on wave-1 anxiety alone.
- *ICU lactate.* In both sets every restricted model is rejected. Lactate is ordered according to pressure and creatinine *jointly*, and the ICU type and ventilation make the fit worse, not better.
- *Diabetes.* HbA1c testing is mostly explained by age (0.094 → 0.016) but not entirely, and readmission and admission route add nothing.

The same models, with admission type and age band as the external covariates, are now part of analysis E of the MIMIC-IV kit. Its synthetic check recovers a planted creatinine-only selection and rejects the other models (9 of 9 checks).

11.3 Structural events against external states

A pipeline (`structural_events.py`) turns any sequence of systems into events: blow-ups and blow-downs (the vertex count of the coupling polytope going up or down, a change of toric type), chamber transitions, breaks appearing or healing, and steepness. It tests their association with an external state by permutation, crude and within strata.

- *ICU deaths* (7,560 stays, day one to day two). No structural event is associated with death, crude or within deciles of the margin-based risk. Blow-ups carry 15.1% mortality against 14.0% ($p = 0.26$, within deciles 0.46); breaks appearing, 13.7% against 14.5% ($p = 0.39$).
- *Partisanship.* Across 22 GSS year-to-year steps, the share of item-pair systems that change chamber correlates with the size of the change in Republican identification at Spearman 0.35 ($p = 0.11$). The direction is suggestive, but 22 steps cannot decide it.

- *Hallucination thresholds.* For the eight retraining tasks we tabulated the regime against the change in incorrect and invalid answers, without a test. The three decided tears are among the four tasks with the largest increase in incorrect answers: HotpotQA, counting happy numbers and primality. The exception is LeetCode, a bend with a 31-point rise.

The pipeline is general. What it lacks is data with many units, a nondegenerate geometry and an external state measured at the same grain.

11.4 A generalized reliability level for the tower

Level three now carries a family of floors. A model's prediction ρ is transported as $\alpha\rho + (1-\alpha)\varphi$, with φ on a grid from 0.05 to 0.40. This is the Kleisli composite of ρ with a binary channel, so the reliability level is a posterior over channels, not over a scalar. It is compared with the fixed floor of section 7.2 and with the evidence-rate floor found there after the fact, which was pre-registered here. Since no new ICU data are open, replication uses four new splits of PhysioNet:

- the sets reversed (R1);
- three shifts between unit types (R2–R4), in both directions of mortality.

On planted data the variants behaved as intended: under a planted prevalence shift both new variants earned their keep, and without one none harmed.

Table 14: The tower's reliability level across arms: gain in test log loss over flat model averaging, with 95% interval, and the predicted death rate. Bold: the interval is above 0.

arm (fit → evidence and test)	death rate (fit / test)	fixed floor	evidence floor	family of floors	predicted (family)
R0a same population, A → B	0.139 / 0.141	0.0000	0.0000	0.0000	0.141
R0b shift, A surgical → B medical	0.102 / 0.184	0.0001	**0.0090** $(0.002, 0.016)$	**0.0125** $(0.005, 0.020)$	0.164
R1 same population, B → A	0.142 / 0.135	0.0008	0.0008	0.0007	0.138
R2 shift, B surgical → A medical	0.106 / 0.165	0.0014	**0.0086** $(0.002, 0.016)$	**0.0111** $(0.001, 0.021)$	0.162
R3 shift, A medical → B surgical	0.173 / 0.105	−0.0004	0.0040 $(-0.001, 0.009)$	**0.0083** $(0.003, 0.013)$	0.125
R4 shift, A coronary/cardiac surgery → B medical/surgical	0.086 / 0.175	0.0012	**0.0093** $(0.002, 0.017)$	**0.0105** $(0.003, 0.019)$	0.176

All three pre-registered hypotheses hold (table 14 and fig. 10):

- The evidence floor replicates in 2 of the 3 new shift arms.
- The family of floors earns its keep in all 3.
- No variant harms in the reversed same-population arm.

The family of floors also calibrates best. It predicts 0.176 against an observed 0.175 in R4, and 0.162 against 0.165 in R2. Its posterior over floors moves where the shift points: towards 0.05 when mortality falls (R3), and towards 0.25–0.40 when it rises. Its Fano tube now agrees with the data's chamber. In R0b, where the fixed floor was certain of the wrong surface ($\mathbb{CP}^2$), the family puts 0.90 on $\mathbb{F}_1$, the surface of the observed system.

The replication is on new splits of the same 8,000 stays, not on new data. Open ICU data with deaths are scarce; eICU and MIMIC-IV are credentialed and would be the external test.

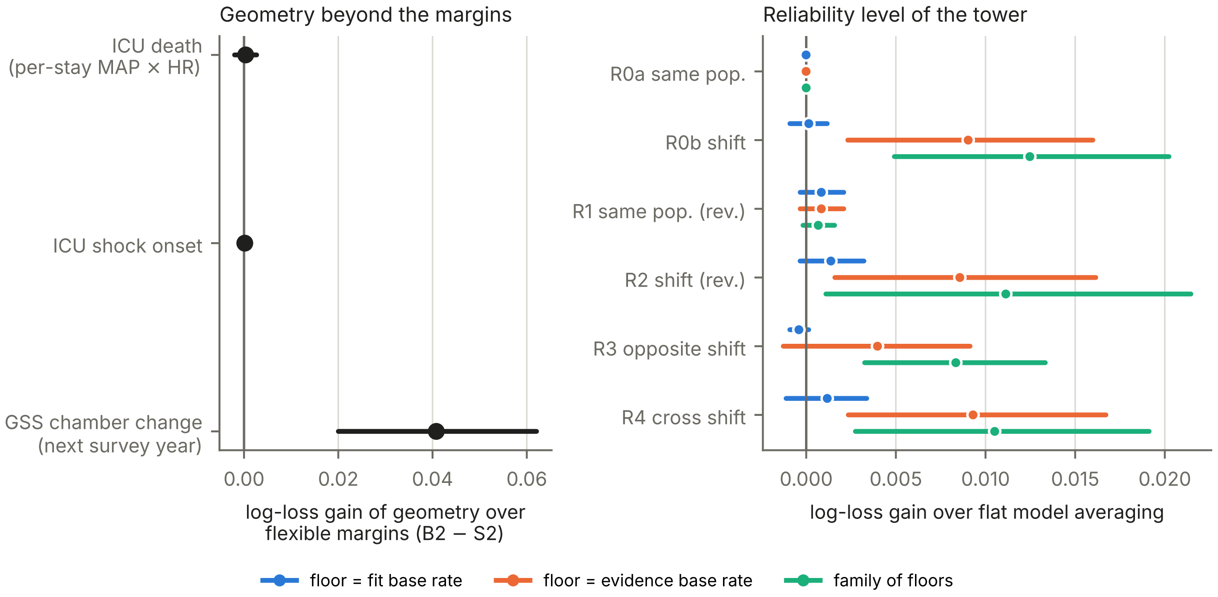


Figure 10: Left: test log-loss gain of geometry over a flexible function of the margins, with 95% intervals, for ICU death, ICU shock onset and GSS chamber changes. Right: gain of each reliability level of the tower over flat model averaging in the six arms of table 14.

12 The context-to-outcome map, and geometry at the boundary

Several systems in this paper carry the same information in several contexts:

- one spending item in two wordings;
- one pressure band in four kinds of unit;
- one question under two prompts.

The natural object is a map from (context, information) to a probability of an outcome, and the natural question is whether it depends on the context. Algebraic geometry states this exactly. With context C, information A and outcome O, *context invariance* $P(O \mid A, C) = P(O \mid A)$ is the conditional-independence model $O \perp C \mid A$. It is a toric variety in the simplex of $C \times A \times O$ tables, cut out by the binomials $p_{cao}\, p_{c'ao'} - p_{cao'}\, p_{c'ao}$. Its distance from the data, the divergence of the information projection, is $I(O; C \mid A)$, and twice the sample size times it (in nats) is the G^2 test. When invariance fails, the binomials say where: at which level of the information the context changes the outcome map. The plan, its planted checks and frozen results are in `ext/phase5`. The planted check held its size (12 rejections in 200) and located a planted change at the right answer level.

GSS: wording, answer and party. Wording is randomized, so here a failure of invariance is a causal effect of the wording on what an answer means politically.

- *Seven items are invariant.* For race, drugs, education, environment, space, defense and foreign aid, the wording changes how people answer but not what their answer says about their party.
- *Four items are not* (Bonferroni over eleven): welfare (mean 0.0126 bits per year), crime, cities and health.
- *Where welfare fails.* The failure sits at “about right” (58% of its G^2). Saying welfare spending is about right is a different political statement from saying assistance to the poor is about right.

The toric decomposition makes this geometric (fig. 11, right).

- The answer × party coupling polytope of welfare has a different toric type in the two wordings in 23 of 24 years: usually 18 vertices for “welfare” and 12 or 15 for “assistance to the poor”.
- In 16 of those 23 years the invariant map reproduces both types. There the change of toric surface is carried by the answer margins alone, as direct influence.
- In the other 7 the context changes the geometry through the outcome map itself.
- No welfare chamber is decided on the bands in both wordings in any year. The toric types are point estimates, while the invariance test, which does not depend on the chamber, is decisive.

ICU: unit type, first-window pressure and death. Invariance fails in both sets ($G^2 = 119$ and 141 on 9 df; 0.022 and 0.026 bits). Its geometry shows how.

- In cardiac surgery the observed death × pressure system is $\mathbb{CP}^2$, while the invariant map would make it $\mathbb{F}_1$. In set B that chamber is decided on the bands.
- In medical units it is the reverse.
- The reason is that the death rate given borderline pressure is 2–3% after cardiac surgery and 21% in medical units. The same reading means different things in different units, and the Fano surface of the unit’s system records it.

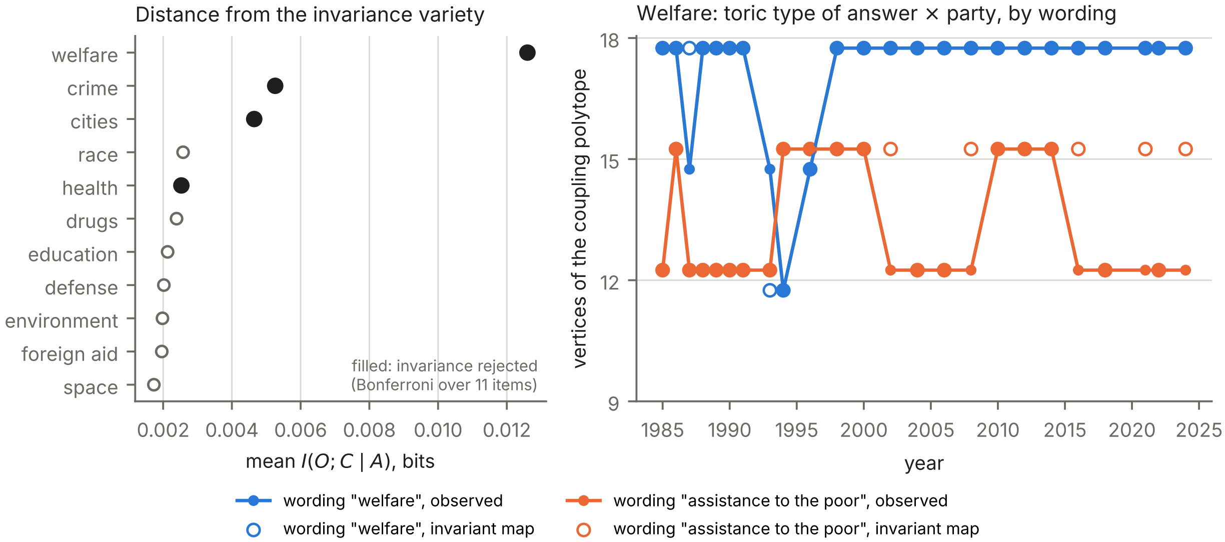


Figure 11: Left: distance of each GSS item’s wording × answer × party table from the invariance variety $O \perp C \mid A$ (mean over 24 years); filled when invariance is rejected. Right: toric type of the welfare answer × party system in each wording, observed (lines) and under the invariant map (circles).

Geometry at the boundary. Per-stay ICU day systems usually have an empty band, and half have a point polytope (section 11.1). A genuine point polytope carries no coupling information: when a margin is a single category the coupling is fixed. That is a theorem, not a failure of the method. But most of these points are not genuine. A band with no hours in 22 is a zero count, not a zero probability. We re-ran the forecast of death with three boundary-aware readings:

- the Dirichlet posterior over toric types;

- the walls decided on the bands, i.e. the family over the band, which is a full-dimensional polytope [7, Prop. 6.1];
- the face stratum, the torus orbit in which a broken system lies.

What the readings recover:

- 71% of day-two systems put more than half their posterior mass on full-support toric types, mostly the 12-vertex type.
- On average 8.7 of the 18 walls are decided.
- The point polygons are therefore largely an artefact of the point estimate, and the geometry behind them is recoverable.

It still does not forecast death beyond a flexible function of the margins: gain -0.0014, interval $(-0.0052, 0.0023)$. The planted check showed why such tests are hard: averaging over the posterior makes the geometric features smooth in the counts, so a flexible margin model approximates them. The first planted check, with a weaker effect, could not detect it; it was replaced before the run by a stronger one matched to the data, which did. So the boundary is not a dead end for the geometry. For this outcome, at this resolution, it is a dead end for prediction.

13 Tracking the approach to collapse: a family of cones, and recursive training

A trajectory of systems traces a path of margins. Over a path, the coupling polytopes form a family whose union is again a polytope [7, Prop. 6.1], and each member's normal fan (a cone at each vertex) changes only where the path meets a wall. The walls $f(I) = g(J)$ form the chamber complex of the marginal map. So the approach to collapse can be read off the path:

- a *bend* keeps the fan;
- a *tear* crosses a wall, and the fan gains or loses a cone: a blow-up or blow-down (for 2×3 systems, of a toric surface);
- a *break* reaches the boundary, and the fan drops dimension.

We built a tracker that reports, at every step of any trajectory:

- the chamber, with signed wall distances and their band intervals;
- tear and break margins, velocity, and tear and break times [7, Def. 3.1, Thm. 3.2, Def. 5.7];
- the drift $D(p_t \| p_{t-1})$, split exactly by Csiszár's Pythagorean identity into *interaction drift* $D(p_t \| p^*)$ and *margin drift* $D(p^* \| p_{t-1})$, with p^* the I-projection of p_{t-1} onto p_t's margins;
- the normal-fan type: dimension, vertices (maximal cones) and facets (rays);
- a Dirichlet posterior over chambers.

Margin drift says who carries which information; interaction drift says what the information means. The plan (`ext/phase6/PREANALYSIS_COLLAPSE.md`) was fixed before anything was computed. Its planted checks passed:

- the Pythagorean identity to 3×10^{-13};
- fan types against the existing toric code;
- a planted wall crossing found at the right step;
- a planted break detected.

Two checks had their specifications corrected before any real run. One rank-tolerance bug, which reported a dimension of 5 for some 3×3 systems, was found in the first trajectory run; it was fixed, a dimension check was added, and all trajectories were rerun.

13.1 Five trajectories

- *GSS,* 110 *item-pair systems over* 40 *years.* Across 2,530 steps the fan is unchanged in 1,824, blows up in 356 and blows down in 350. Only 18 steps carry a decided chamber change. The run-up to the decided welfare × race tear (wording Y, 2014→2016) is visible in advance (fig. 12, top left):
    - the tear margin shrinks from 0.018 (2010) to 0.013 (2012) and 0.006 (2014);
    - the tear time falls to 0.09 steps, the smallest since 2008, when the path passed within 0.0002 of a wall and the point fan dipped to 12 vertices without any decided change;
    - the posterior probability of the modal chamber drops to 0.48 in 2014;
    - interaction drift peaks in 2012 (0.031 bits, against a median of 0.006);
    - after the tear the chamber is decided (0.99) and the fan has lost three cones (15 → 12).
- *C19PRC waves.* Depression × anxiety keeps the same 15-vertex fan through all waves in both countries. It sits very close to a wall (UK tear margin 0.0005–0.013) without a decided change.
- *The tower's evidence paths.* Hypothesis change is motion through chambers. In the two shift arms with a mortality rise (R0b, R2), the level-one prediction crosses the wall $f_1 = g_0$ twice as evidence accrues, and the fan alternates between $\mathbb{CP}^2$ and $\mathbb{F}_1$. In the same-population arm and in R4 it stays in one chamber.
- *ICU decedents against survivors.* In 6-hour windows over 48 hours, the decedents' pressure × heart-rate system hugs a wall after the first window: tear margin 0.001–0.018, against 0.020–0.046 for survivors. It keeps a 15-vertex fan, where the survivors' drops to 12 after the first window. Neither group approaches the boundary (break margins about 0.04–0.06). The two groups, the same information in two outcome contexts, differ by 0.05–0.08 bits per window, of which 88–98% is margin drift.
- *LLM retraining (two points).*
    - Happy numbers and ARC *break*: the fan drops dimension because a category empties.
    - HotpotQA *blows up* (9 → 12 vertices).
    - Primes makes four decided wall crossings with the same fan.
    - In every task the drift is overwhelmingly in the margins (up to 3.4 bits), with interaction drift 0.003–0.04 bits.

The forecasting test. Does interaction drift, or tear time, forecast the next *decided* chamber change beyond plain margin velocity? The test set has only 11 such events, below the pre-registered minimum of 15. The test is underpowered, and neither predictor adds anything: gains -0.0001, with intervals spanning 0. On the secondary target, point-estimate chamber changes (634 test events):

- tear time adds 0.017, with an interval $(-0.001, 0.033)$ just touching 0;
- tear time with interaction drift adds 0.017, $(0.0002, 0.034)$;
- interaction drift alone adds nothing.

So distance to the walls carries the forecast, while the meaning-change part of the drift does not, at least not for the GSS.

13.2 Recursive training with known truth

Design. A model of an outcome y (eight values) given information x (six values) in three contexts was retrained on its own samples for 20 generations: 200 samples per cell, 50 replicates. The truth satisfies context invariance, and some outcome categories are rare tails. Fresh real data were mixed in at 0, 5, 10 and 25%; a variant passed each generation through a paraphrase kernel. Five

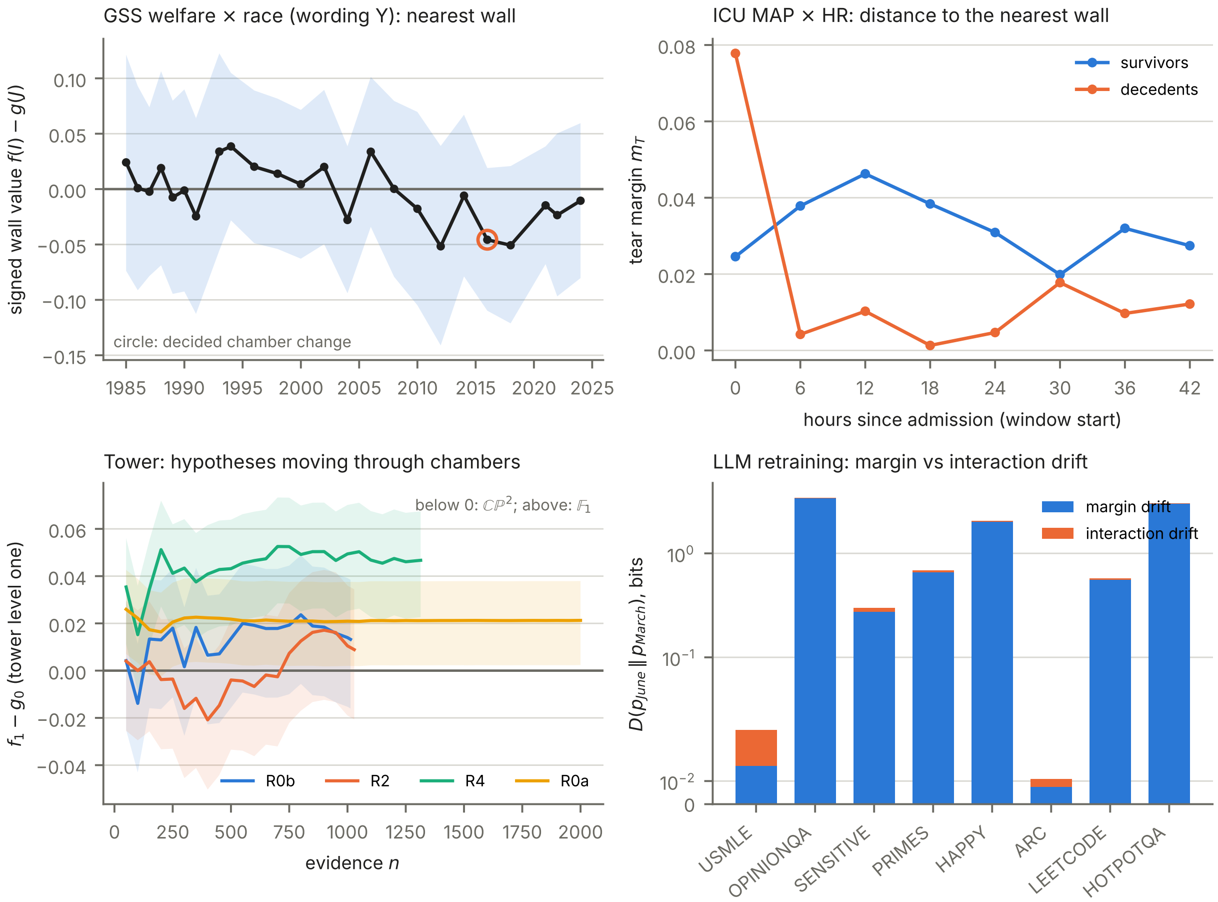


Figure 12: The tracker on four trajectories. Top left: the GSS welfare × race system (wording Y), signed value of the wall nearest in 2014 with its band interval; the circle marks the decided chamber change. Top right: tear margin of ICU pressure × heart-rate systems in 6-hour windows, decedents and survivors. Bottom left: the tower's level-one paths relative to the wall $f_1 = g_0$, with the interval from the pressure band. Bottom right: the March-to-June drift of the LLM systems split into margin and interaction parts (bits, symmetric log scale).

predictions were fixed in advance; two hold and three fail (fig. 13).

1. *Contraction: holds, but trivially.* The paraphrase kernel's Dobrushin coefficient is 1: it shifts outcomes to neighbours without mixing distant ones. The planned bound therefore reduced to "no increase beyond noise". The model's distinguishability of x falls from 0.70 to 0.24 over 20 generations, inside an exploratory sharper bound from the coefficients of the composite kernels K^g.

2. *Breaks in finite time, forecast by break time: fails.*
    - Without fresh data, 88.5% of tail categories are absorbed at zero by generation 20, just short of the planned 90%; 36% are zero already in the model fitted to real data, a break caused by finite data alone.
    - Break time correlates with the remaining time to the break (Spearman 0.33), but the level alone does better (0.48).
    - Given the level, break time adds nothing (0.03, $p = 0.065$).

3. *Fresh data prevents absorbing breaks: fails as stated.*
    - Bands Prop. 9.2 guarantees a floor λP_0 for the training law, not for a model fitted to 200 samples.
    - Categories rarer than about $1/(\lambda \cdot 200)$ are zero most of the time whatever λ is: 29–38% of tracks have three consecutive zeros with fresh data, against 44% without.
    - Exploratory: the qualitative claim holds. Without fresh data a zero never recovers; with fresh data zeros recover at 7–19% per generation, and persistent zeros fall from 37% to 16, 11 and 5%.

4. *Contexts drift off the invariance variety: holds, decisively.*
    - Without fresh data, the model's $I(Y; C \mid X)$, its distance from the toric variety $Y \perp C \mid X$, grows steadily from 0.016 to 0.18 bits, in every replicate.
    - With fresh data it plateaus at 0.09, 0.058 and 0.03 bits for 5, 10 and 25%.
    - Retraining contexts on their own outputs makes the same information mean different things in different contexts. That is the "bogus information" of the question, measured exactly, and fresh data cap it.

5. *Certificates warn before accuracy drops: fails.*
    - The tower-style certificate against held-out real data (the distance to the held-out band plus the Cromwell mass the model has zeroed) rises steadily from the first generation.
    - But accuracy also falls from the first generation, and its first 5-point drop comes at a median generation of 2 (range 1–10).
    - The certificate crosses its null threshold strictly first in 58% of replicates, short of the planned 80%; the median lead is 2 generations.

What this shows. The tracker turns any trajectory into a sequence of chambers, fans and drifts, and on real data it shows approach to a wall before a tear. It did so for welfare × race, for the decedents' physiology, and for the tower's hypotheses. In recursive training, the clearest signal of information being destroyed is not the loss of tails, which finite data cause from the start. It is the steady departure of the contexts from the invariance variety. That departure has no counterpart in accuracy and is invisible to any single-context metric, and fresh data bound it in proportion to their share. What did not work as predicted:

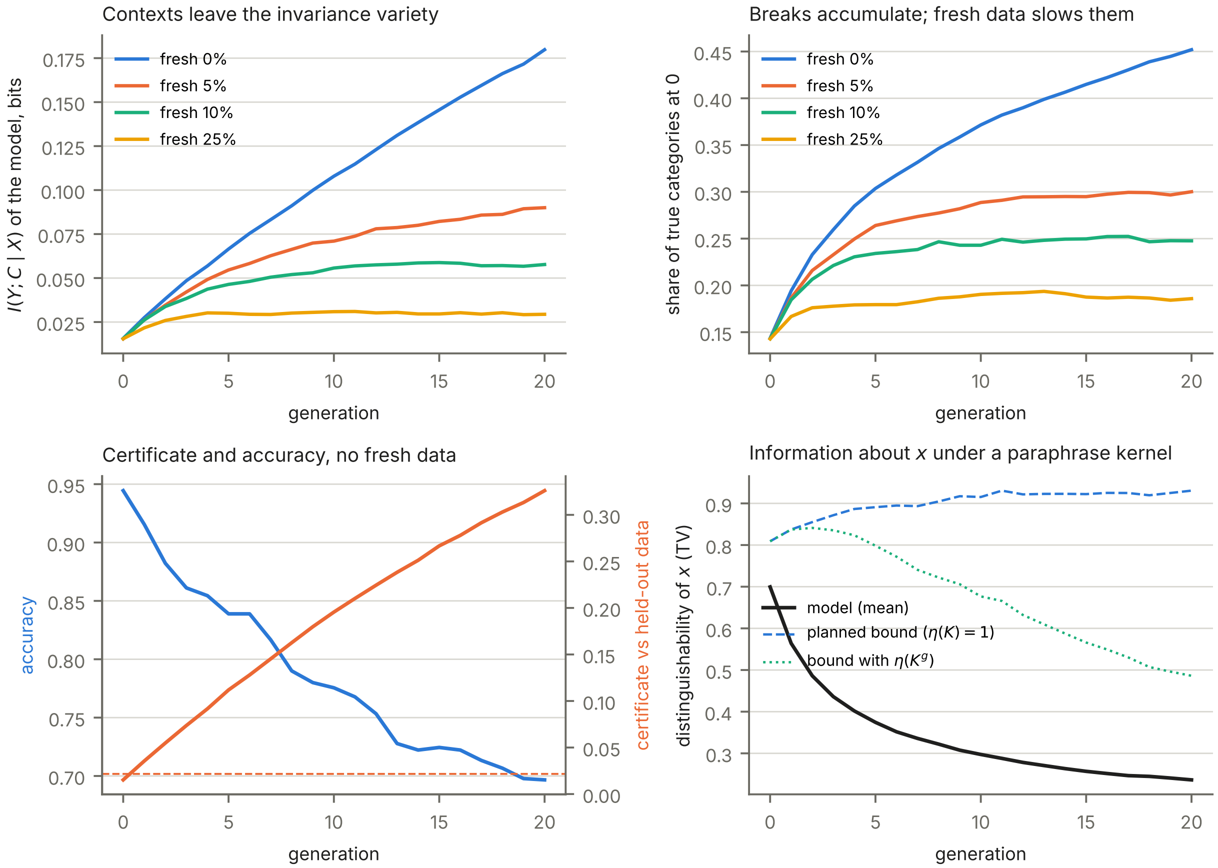


Figure 13: Recursive training with known truth (50 replicates). Top left: the model's distance from the context-invariance variety, $I(Y;C \mid X)$, by fresh-data share. Top right: share of true outcome categories the model has at zero. Bottom left: mean accuracy and mean certificate against held-out real data without fresh data; the dashed line is the certificate's null threshold. Bottom right: distinguishability of the information under a paraphrase kernel, with the planned and the sharper bound.

- break time as a forecaster beyond the level;
- Prop. 9.2 as a finite-sample floor;
- the certificate as an early warning with a large lead.

Each needs a sharper statement or a longer horizon, and the analysis does not go beyond these results.

14 Ambient support geometry: deciding under uncertainty with floors, typed obstructions and replayable records

The tracker of section 13 describes; a decision has to be made from it. This section integrates the proposal *Ambient Support Geometry: Presheaves, Positive Support Floors, and Local-to-Global Obstructions* [41] into the programme, sharpens it where the data and the mathematics allow, replaces its comonadic layer by a fibred, locally posetal 2-category, and reports a pre-registered test of its experimental programme (`ext/phase7`, plan `PREANALYSIS_SUPPORT.md`, hash frozen before any experiment; code hashed before the run).

14.1 Support as a policy, and the local-first principle

A probability model normally lets a coordinate approach zero. In applications the question is often not whether an outcome is possible, but whether it has enough support to be relied on: a rare event may be possible and still too weakly supported for a particular decision, while a strict floor can be wrong when rare but real outcomes must be kept. The threshold therefore belongs to the ambient decision context, not to probability alone. Following the proposal, a decision combines four pieces of information:

1. a *support policy*: which outcomes or events must have probability at least a chosen floor;
2. the *local probability geometry*: margins, conditionals and coupling constraints;
3. *provenance*: which observations, assumptions and policy choices led to the decision;
4. an *obstruction signature*: the weakest support margin, and any local-to-global or composition failure.

The aim is not to force every local structure into a global model, but to make each local conclusion auditable and to expose where the chain from evidence to outcome becomes fragile. A global emergent object may not exist, or may exist only under assumptions the evidence does not justify; so a global model is not the default data structure. Each context carries its own local probability object, policy and evidence trail; restriction says what remains visible in a smaller context; compatibility asks whether local objects agree on overlaps; gluing asks whether compatible local objects come from one global object. This is the use-case for a presheaf $F : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ (or to constraint objects or records), with restriction maps $\rho^U_V : F(U) \to F(V)$ for $V \subseteq U$. A presheaf asserts no global section; in an AMB measurement scenario a compatible family that fails to extend is a contextuality witness, meaningful precisely because local compatibility does not force a global section [39].

14.2 Floors and the coupling polytope

For margins $f \in \Delta(X)$, $g \in \Delta(Y)$ and floors $\tau \geq 0$ on a set D of designated cells, the support-constrained polytope is

$$P_\tau(f, g) = \{\pi \in P(f, g) : \pi_{xy} \geq \tau_{xy} \text{ for } (x, y) \in D\}; \tag{1}$$

more generally one may floor events, $\pi(E_k) \geq \tau_k$.

Remark 14.1 (Two thresholds). A floor has probability units, $0 \le \tau \le 1$. A safety factor such as "1.1" is dimensionless: relative to a baseline floor $\tau^{(0)}$ it means $\tau \ge 1.1\,\tau^{(0)}$; it is not a probability of 1.1. The decision layer below also uses a *relative* threshold κ between competing actions, $q_x \ge \kappa q_y$: the wall $q_x = q_y$ replaced by the binomial $q_x = \kappa q_y$. A multiplicative constant keeps the binomial form (a translate of the toric variety); an additive one, $p^u - p^v = c$, does not. Both thresholds are implemented and kept apart.

Proposition 14.2 (Feasibility is policy-relative [41])**.** *$P_\tau(f,g)$ is a possibly empty polytope. If it is empty, the margins and the policy cannot hold together; this does not by itself make the data false: the floor may be too strong, the margins uncertain, or the cells unsuitable for a common guarantee.*

The proposal cautions that truncation "need not preserve the original toric fan". For floors on cells it does, exactly:

Proposition 14.3. *Let $f' = f - \tau\mathbf{1}_D\mathbf{1}$ and $g' = g - \mathbf{1}^\top\tau\mathbf{1}_D$ be the reduced margins. Then $\pi \mapsto \pi - \tau\mathbf{1}_D$ is a bijection from $P_\tau(f,g)$ onto $P(f',g')$. Hence P_τ is non-empty iff $f', g' \ge 0$; its normal fan is that of $P(f',g')$; its walls are $f'(I) = g'(J)$; and the largest common margin above the floors with weights w,*

$$\delta^\star = \max\{\delta : \pi \in P(f,g),\ \pi_{xy} \ge \tau_{xy} + w_{xy}\delta \text{ on } D\}, \tag{2}$$

equals min *over rows and columns of (margin − floors)/(weights) whenever that value is non-negative, with a row or a column as the bottleneck.*

Proof. On D, $\pi \ge \tau$ is $\pi - \tau\mathbf{1}_D \ge 0$; off D the shift is zero; the margins shift by the row and column sums of $\tau\mathbf{1}_D$. A transportation polytope with non-negative margins of equal total is non-empty. For (2) apply the same shift with $\tau + w\delta$; it is valid while $\tau + w\delta \ge 0$. ☐

So a floor alone can cross a wall: a policy can tear a system. On the uniform 2 × 3 hexagon a floor on one cell crosses two walls at $\tau = \frac{1}{6}$ and breaks the system at $\tau = \frac{1}{3}$ [9, Ex. 6.10]. Floors on events give general polytopes, and the proposal's caution applies to them. The support obstruction for cell floors needs no optimisation: its certificate reads "this row carries f_x, but its floors demand more"; with margins known only to their bands, a floor is decided-feasible when the lower band ends cover the floor sums. The weakest link is kept as a typed signature, not a single score,

$$\mathfrak{O} = (\mathfrak{O}_{\mathrm{supp}}, \mathfrak{O}_{\mathrm{glue}}, \mathfrak{O}_{\mathrm{comp}}, \mathfrak{O}_{\mathrm{evid}}),$$

recording support shortfall, failure to glue, failure to compose and inadequate or conflicting evidence. Its components have different mathematical types, and combining them requires an explicit normalisation and decision policy.

14.3 Chains, presheaves, and two vocabularies kept apart

The proposal's "spring leaf" is a *local constraint chain*: a path of contexts $U_0 \leftarrow U_1 \leftarrow \cdots \leftarrow U_n$ with local states $s_i \in F(U_i)$, restriction or transport maps between them, and a support slack and obstruction recorded at each stage. A conclusion is carried forward only while the compatibility conditions hold; when they fail, the chain records where and why, instead of inventing a global state. A chain follows one route; a presheaf records all routes, and two routes that transport a conclusion differently give a path-dependence (holonomy) diagnostic [45].

Two vocabularies are used side by side. On the AMB side [39], a measurement scenario has contexts and an assignment presheaf, and a compatible family that fails to extend is a contextuality witness. On Vigneaux's side [42, 43], an information structure has observables with a conditioning action, and information functionals such as entropy live there. The support ambience attaches floors, bands, provenance and obstruction records to each context, which enriches the local data but does not identify the two coefficient systems. In particular the support indicator (positive section $\mapsto 1$, empty $\mapsto 0$) is not additive, so it is not a morphism of abelian coefficient systems and induces no map of cohomologies; nor does a holonomy cocycle of known transformations become the AMB obstruction because both are local. The division of labour is: the presheaf says what local states exist and how they restrict; the floored polytope, which assignments meet the constraints; the conditioning action, how information functionals respond to conditioning; the AMB extension test, whether compatible support sections extend [48]; provenance, which evidence and policies produced the result.

14.4 Carrying the ambience: a fibred, locally posetal 2-category

The proposal carries the ambient record $a = (m, \tau, \mathcal{E}, \mathcal{G}, \mathcal{D})$ (constraints, floors, evidence, gluing rules, decision policy) by an informal comonad $C_a(Y) = Y \times A_a$, and leaves its laws and its transport rule open. The programme already carries the tower's context in the way this asks for. Over each finite base B the context sits in a reader comonad $\mathsf{C}^B(Y) = Y \times A_B$; every Markov kernel k transports it by a comonad morphism $\mathrm{id} \times A_k$, functorially on the Kleisli category [34, Prop. 9.14]; and the category of elements of $B \mapsto A_B^{\mathrm{coh}}$ is a split opfibration over the Kleisli category, along whose lifts the comonads are carried [9, Prop. 3.3], certificates pull back and failure levels can only rise [9, Thm. 3.1]. So the proposal's comonad, read correctly, is that family: one comonad per context, carried along transports. What the ambience adds is *order*: the fibres of the existing opfibration are sets of points (coherent contexts), and a support policy needs sets of laws ordered by strength. The construction below is the posetal enlargement of the existing one; its laws are immediate, and its content is the transport rule.

Definition 14.4 (Ambience functor and its total category)**.** Let $\mathcal{C}$ have contexts as objects and transports as morphisms $t : U \to V$: marginalisations (restriction to fewer variables), deterministic maps and Markov kernels K_t on cells. Let $\mathcal{A}(U)$ be the poset of closed convex feasible sets $S \subseteq \Delta(X_U)$ (credal sets, floored polytopes), each carrying its record, ordered by inclusion: $S \leq S'$ if $S \subseteq S'$, so an arrow is a *weakening*. Transport acts by images, $t_! S = K_t S$. The ambience's total category $\int \mathcal{A}$ (its Grothendieck construction [47, 46]) has objects (U, S) and an arrow $(U, S) \to (V, S')$ for each $t : U \to V$ with $t_! S \subseteq S'$. The projection $\pi : \int \mathcal{A} \to \mathcal{C}$ is an opfibration whose fibres are posets. Restricted to kernels and to singletons $S = \{\rho\}$ at level one, it is the level-one part of the discrete opfibration of [9, Prop. 3.3].

Proposition 14.5. *1. Images compose strictly, $(st)_! = s_! t_!$, so $\mathcal{A} : \mathcal{C} \to$* **Pos** *is a functor. A decision is a path in $\int \mathcal{A}$; its replay is the composite, which depends only on the recorded path.*

2. Imposing a policy, $\iota_\tau(S) = S \cap \{\pi \geq \tau\}$, is an idempotent comonad on each fibre (an interior operator: $\iota_\tau S \subseteq S$, $\iota_\tau \iota_\tau = \iota_\tau$); summarising a set by its floors, $\sigma(S) = \{\pi \in \Delta : \pi \geq \min_S \pi\}$, is an idempotent monad (a closure operator: $S \subseteq \sigma S$, $\sigma\sigma = \sigma$). The counit, comultiplication and unit laws hold trivially in a poset.

3. Neither commutes with transport, and the failures point one way: $t_! \, \iota_\tau S \subseteq t_! S \cap \{\pi \geq K_t \tau\}$ and $\sigma \, s_! \, \sigma \, t_! S \supseteq \sigma \, (st)_! S$. Imposing a policy before transport certifies at least as much as pushing the floor forward; summarising between steps certifies at most as much as carrying the set. Each inclusion is a 2-cell: on floor boxes, $t \mapsto \sigma \circ t_!$ is a normal oplax functor into posets [34, Prop. 10.24].

4. *On a discrete (op)fibration (a* **Set***-valued ambience, whose fibres have only identity arrows) every fibred comonad is the identity: a counit* $c(a) \to a$ *inside a discrete fibre forces* $c(a) = a$*. This is why the programme carries comonads* alongside *its discrete opfibration, as a family* C^B *joined by comonad morphisms, and not* on *it.*

Proof. (1) $K_s K_t = K_{st}$, and the composite of a path is determined by its arrows. (2) Intersection with a fixed set and taking the floor box are deflationary-idempotent and inflationary-idempotent; in a poset every such operator is a comonad, respectively a monad. (3) If $\pi \in S$ and $\pi \geq \tau$ then $K_t\pi \in K_t S$ and $K_t\pi \geq K_t\tau$, since $K_t \geq 0$; and $\sigma t_! S \supseteq t_! S$ gives $s_!\sigma t_! S \supseteq (st)_! S$, then apply the monotone σ. (4) Immediate. ☐

So the structure is a 2-category fibred over the contexts and *locally posetal*: 0-cells are contexts with their feasible sets, 1-cells are transport laws (the image $t_!$, the naive pushed floor $K_t\tau$, the tight floor $\min_{t_! S}$), and 2-cells are inclusions between them. The naive floor sits below the tight one: in the pre-registered transport experiment (E5 below) the two agree for only 27% of target cells. In an exploratory run of 300 random two-step chains (`lax_composition.py`) the stepwise-summarised floor fell strictly below the composite floor in 611 of 895 target cells and never above it, and imposing the policy first certified strictly more than the transported floors combined with the pushed floor (their componentwise maximum) in all 1,158 cells. The consequence for the weakest link is precise: carrying sets, transport loses nothing; certification is lost where a set is summarised by floors, or where a policy is pushed forward instead of imposed. Along a whole leaf-spring chain of transports and vertical steps this becomes a theorem [34, Thm. 10.26]: certification is path-independent, it is lost only at summaries and weakenings, the loss splits exactly into non-negative contributions, one per summary, and imposing early and summarising late certifies most. The proposal's intuition of a comonadic layer survives in this modest form, policy imposition as an interior operator on each fibre, while its open obligation, the transport law, becomes the oplax comparison of proposition 14.5(3), which can be computed. The ordered total category and its floors are now part of the theory papers [9, Def. 3.5, Prop. 3.6], [34, Section 10.7]. The ambience is deliberately a *prestack*, not a stack: descent is not imposed, because the failure of descent is the information.

Why not other ways. Each alternative below was considered and rejected for the stated reason.

Alternative	Why not, or what it is good for
Product (reader) comonad $Y \times A$	Used, not rejected: it is the tower's per-base comonad C^B, carried along kernels by comonad morphisms [34, Prop. 9.14]. With one fixed parameter, as in the proposal's $Y \times A_a$, it cannot follow a change of context and keeps no history.
Fibred comonad on $\int A$, A set-valued	Trivial by proposition 14.5(4): discrete fibres admit no non-identity counit; hence the family of comonads above.
A general (non-posetal) 2-category, natural transformations as 2-cells	Coherence obligations without content: the fibres are posets of feasible sets, so 2-cells are inclusions and every diagram of 2-cells commutes. The quantitative counterpart is the graded shadow of [6, Thm. 7.8], a strict 2-category enriched in Lawvere metric spaces; with couplings as 2-cells the witness layer is neither a 2-category [6, Thm. 7.6] nor a double category, strict or lax [3]. The oplax comparison here is of a functor into posets, not of those layers.
Sheaf or stack	Imposes descent, i.e. that compatible families glue; here the failure to glue is the datum to be recorded.
Topos and internal logic	Not needed: support is graded by headroom, not by truth values, and nothing uses a subobject classifier.
A single global model (Bayesian network, global joint)	Presupposes the global section whose existence is in question.
Writer monad (an append-only log)	Records history but not what restriction means; it is how the records are stored (`support.record`), not the mathematics.
Markov category [44]	The natural home for the 1-cells (composing kernels); it carries neither policies nor gluing data, and can serve as $\mathcal{C}$.
A scalar confidence score	Flattens the four obstruction types; a shortfall, a Farkas vector, a holonomy and a lying source are different evidence.
A cohomology map AMB → Vigneaux	Blocked: the support indicator is not additive (above).

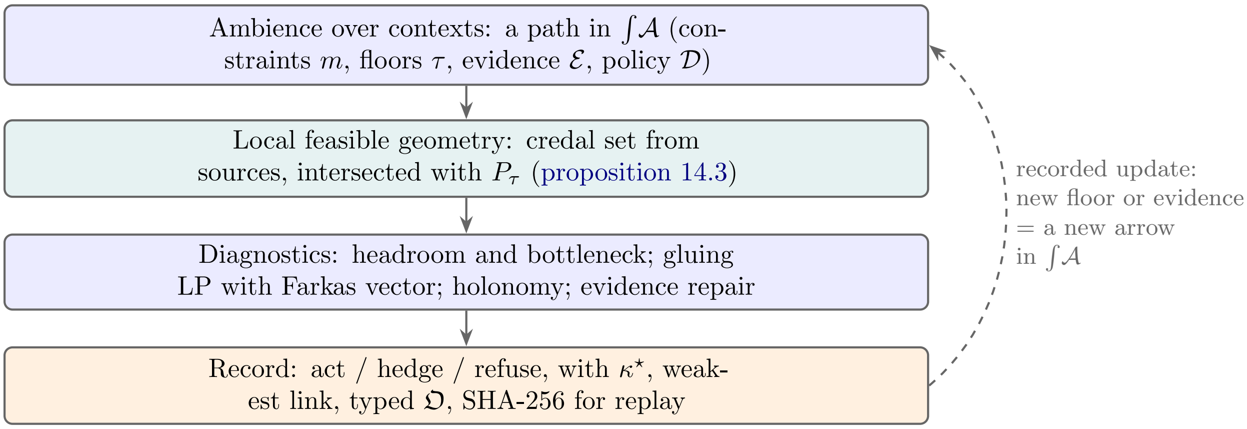

Ambience over contexts: a path in $\int\mathcal{A}$ (constraints m, floors τ, evidence $\mathcal{E}$, policy $\mathcal{D}$)
Local feasible geometry: credal set from sources, intersected with P_τ (proposition 14.3)
Diagnostics: headroom and bottleneck; gluing LP with Farkas vector; holonomy; evidence repair
Record: act / hedge / refuse, with $\kappa^\star$, weakest link, typed $\mathfrak{D}$, SHA-256 for replay
recorded update: new floor or evidence = a new arrow in $\int\mathcal{A}$

Figure 14: The decision architecture of [41], with its comonadic layer replaced by the total category of the ambience. Feedback changes the ambience only through a recorded arrow, so every decision replays from its path.

14.5 The decision layer

Sources, reliabilities and the credal set. Each source c reports counts over its own outcomes, with a reliability r_c (or an interval, whose low end is used) and, where its context legitimately changes the outcomes, a known map T_c. Writing $P_c = r_c T_c q + (1 - r_c) F_c$ with F_c unrestricted, the band of P_c bounds the true outcome law q from above and below; all sources together give a polytope Q, intersected with the policy's floors. When the sources contradict each other Q is empty, a regional failure to glue: an elastic linear program names the conflicting sources and outcomes, every single exclusion that would restore consistency is listed, and by default the least reliable of these is excluded, with the largest reliability its data allow recorded ($\mathfrak{O}_{\text{evid}}$). A floor the sources cannot support is reported, with its shortfall and binding bounds, not imposed ($\mathfrak{O}_{\text{supp}}$). Each decision is one linear program per pair of actions; the output carries the strength κ^*, a trace that splits the worst-case margin exactly among the sources by duality, the binding bounds, the weakest link (for each source, the reliability below which the decision fails), the thinnest cell, and which source's further data would help most.

A worked example (synthetic). Three answers a, b, c; four sources: a reference text ($r = 0.9$), a forum ($r = 0.3$, whose falsehoods push c), a paraphrase set ($r \in [0.6, 0.7]$, pushing b), and self-generated text claiming $r = 0.8$ but carrying no true signal. The self-generated source is found to contradict the others (its data allow at most $r = 0.03$) and is excluded. "Assert a" is then supported up to $\kappa^* = 1.68$: the layer acts at $\kappa = 1.1$ and 1.5 and hedges between a and b at $\kappa = 2$. The decision does not depend on the reference text at all; it rests on the forum's upper bound on b (the forum is mostly wrong but rarely says b), so its weakest link is the forum, whose reliability must stay above 0.235 (stated 0.3) at $\kappa = 1.1$.

A decision record. The proposal's illustrative record, filled in from a real run (one register, $r = 1$, counts 600/300/100 over a, b, c; policy: c at least 0.30):

Field	Record
Decision	Support obstruction: no action
Policy	$q_c \geq 0.30$
Active constraint	the floor on c (bottleneck)
Witness	the register's band forces $q_c \leq 0.125$ (dual -1 on its upper bound for c)
Weakest link	shortfall 0.175
Provenance	source counts, reliabilities and policy version, hashed
Next action	revise the floor, revise the evidence, or defer

With the floor at 0.05 instead, the layer acts ("assert a") with headroom 0.075 on the floor, the same bound binding.

14.6 Entropy, Fisher geometry and the three gluing statuses

A floor changes the admissible region; it does not turn a toric equation into one with a positive constant (remark 14.1). On the interior of the simplex, entropy and the Fisher metric are regular, and a floor keeps the floored coordinates away from the boundary, which can only improve regularity on the constrained region; by proposition 14.3 cell floors keep the toric type of the reduced margins, while the Fano property and the disc potential still have to be checked for each new polytope. For gluing, with (X, Y) and (Y, Z) margins fixed, every filler satisfies $H(X, Y, Z) \leq H(X, Y) + H(Y, Z) - H(Y)$, with equality exactly when $X \perp Z \mid Y$. A floor can exclude this maximum-entropy filler while other fillers survive, so three statuses are kept apart: a filler exists; a filler meets the policy; the maximum-entropy filler is among the admissible ones.

14.7 The pre-registered programme

The proposal's minimal experimental programme was pre-registered and run once; two checks failed as specified and are reported with their corrections.

- *E1, floored geometry.* In 500 random floored systems (2×2 to 3×4, random designated cells), the vertices, enumerated independently, are the shifted vertices of the reduced-margin polytope in all 500 (worst discrepancy 10^{-8}), and the 2×3 toric surfaces agree in 84 of 84; all 200 systems with a negative reduced margin are infeasible. As run, the check reported 324 of 500, because its comparator sorted rounded vertex lists, which is order-sensitive; the order-free comparison of the same instances is a reported deviation.
- *E2, weakest link.* The closed form of proposition 14.3 equals the linear program in all 418 cases with headroom. The planned check on the duals failed (82% against 99%): duals on margin equalities are defined only up to adding a constant to the rows and subtracting it from the columns. The gauge-invariant version (the binding floors lie in the bottleneck row or column) holds in 418 of 418. Without headroom the closed form does not apply; it differs in 43% of cases.
- *E3, the chain $X - Y - Z$* (200 binary instances, 20 floors each). A floored filler exists iff $\tau \leq \min \pi/2$ over the two given tables (4,000 of 4,000). The conditional-independence filler meets the floor iff $\tau \leq \tau_{CI} = \min_{x,y,z} \pi(x,y)\pi(y,z)/\pi(y)$, always the smaller threshold (200 of 200), and the maximum entropy over admissible fillers attains the bound exactly when $\tau \leq \tau_{CI}$ (2,600 of 2,600). Between the two thresholds a floor-meeting filler exists, but not the least committal one.
- *E4, the cyclic cover $(X,Y), (Y,Z), (Z,X)$.* Gluability by linear program agrees with the 3-cycle inequalities [40] in 600 of 600 families; all 81 contextual families come with a Farkas vector, verified independently. Of the 519 gluable families, 124 (24%) are *support-contextual*: every local table meets its floor, the family glues, and yet no global section meets the floor. This is a new obstruction type, a recorded witness rather than a cohomology class. For the binary triangle it has a closed form: a floored global section exists iff every pairwise cell is at least 2τ and the 3-cycle inequalities hold with right side $1 - 8\tau$ [9, Thm. 4.8], so support-contextuality is the band between those inequalities at τ and at 0; the triangle with cells 000 and 111 empty and the rest $\frac{1}{6}$ is support-contextual at every $\tau \in (0, \frac{1}{12}]$ [9, Ex. 4.9].
- *E5, transport.* Floors push forward through any non-negative kernel: $\pi \geq \tau$ implies $K\pi \geq K\tau$ (500 of 500). Under deterministic coarsening the slack of a state never shrinks (250 of 250); under a stochastic kernel it stays above the smallest row sum of K times the original slack (250 of 250) and shrinks in 2% of cases. The pushed floor is the tight one for 27% of target cells: the 2-cell of proposition 14.5 is usually not an identity.
- *E6, records.* All 400 recorded decisions (100 problems, each through three changes of floor, evidence and κ) replay bit-identically from their records, and 20 planted obstructions of each type are classified to that type alone (80 of 80).

14.8 Real retrieval data: SciFact

SciFact [49] pairs 1,409 expert-written scientific claims with 5,183 abstracts, labelled SUPPORT, CONTRADICT or not enough information (NEI) per claim–abstract pair. We used oracle abstracts: 919 training pairs and 340 dev pairs (gold 138/71/131); the unlabelled test split was not used. The sources are three weak classical verifiers trained on the training split. Pretrained inference models could not be loaded here, so the sources are much weaker than modern ones, and every result below should be read as a test of the decision layer under weak evidence. Two pre-registered analyses were run; the first comes from an earlier branch of this work, and the second was planned without knowledge of the first's results.

Analysis 1: channels and credal rules (`ext/scifact`, plan frozen before any model was fitted). Each verifier's confusion channel, and their joint channel over the 27 output patterns, were estimated by five-fold cross-validation on the training split, with bands; credal rules assert a label when its lower posterior beats every rival's upper posterior by $\kappa = 1.1$.

- Baselines: best single verifier 64.7% accurate, majority vote 60.9%, naive Bayes 66.8% (macro-F1 0.61).
- The credal rule on the empirical joint channel acts on 21% of pairs and is 88.7% accurate when it acts, against naive Bayes's 66.8% on all pairs: *H1 holds* (difference 0.22, interval 0.15–0.29). The rule that assumes independent errors acts on 34% at 86.2%; the rule with no dependence assumption (Fréchet bounds) never acts.
- *H2 fails*: the pairs on which the independence rule acts but the joint-channel rule does not were not measurably worse (difference 0.07, interval -0.07–0.21, 45 pairs).
- *H3 holds*: the strength κ^* ranks correct above incorrect labels (AUC 0.59, interval 0.53–0.66).
- *H4 fails*, against the geometry: ranked by κ^*, the area under the risk–coverage curve is *larger* than when ranked by naive Bayes's confidence (0.249 against 0.192, interval of the difference -0.088 to -0.028).
- Exploratory: at the same coverage the credal gate picks the same pairs as confidence thresholding (71 of 71), and it only ever asserts NEI. Its hedges contain the gold label in 95% (SUPPORT or NEI), 100% (all three) and 91% (SUPPORT or CONTRADICT) of cases. No dev claim with several abstracts produced a decided SUPPORT on one and a decided CONTRADICT on another.

Analysis 2: conformal reliabilities and the decision layer (`ext/scifact/conformal`). Here each verifier's reliability is the conformal quantile of the probability it gives the true label on a held-out calibration quarter of the training split, at level 0.05/3. Then the credal set of section 14 contains the truth whenever every source covers it, and acting on a wrong label is impossible at $\kappa \geq 1$, so $P(\text{act and wrong}) \leq 0.05$: a finite-sample guarantee, under exchangeability, that needs no model of how the verifiers' errors depend on each other.

- *P1 holds*: no wrong act on dev; some source missed the truth on 3.5% of pairs, within the 5% bound.
- *P2 fails*, decisively: the reliabilities are 0.056, 0.039 and 0, the credal set is nearly the whole simplex, and the layer acts on 4 of 340 pairs (1.2%, all NEI, all correct).
- *P3 holds*: three sources act on 4 pairs at the same guarantee where the best single source acts on 1, and that one is wrong. *P4 holds* only formally (4 pairs; interval 0–0.5). *P5 holds*: the best verifier is 47 points more accurate on the pairs the layer acts on than on those it hedges (interval 0.41–0.52).
- Risk–coverage (an amendment fixed before the dev run, after a dry run on training pairs): at total levels 0.05, 0.10, 0.20 and 0.30 the layer acts on 1%, 2%, 9% and 16% of pairs, and its act-and-wrong rate (0, 0.003, 0.038, 0.059) stays inside every guarantee. Its accuracy when acting falls to 57% at level 0.20: the guarantee bounds wrong acts over all pairs, not the error rate among acts.

A disclosure: when the second plan was written, its author did not know of the first analysis, and its plan file overwrote the first's in the project folder. The original was restored byte-identical from the session record and all of its frozen hashes verify; the second analysis's files were moved to their own folder, with only two file paths changed before its single run.

What SciFact says. With weak sources, the decision layer behaves as designed and is honest about it: it is right when it acts, it hedges on the hard pairs, its hedges usually contain the truth, and with conformal reliabilities its wrong acts stay within a guarantee that assumes nothing about dependence. It is also nearly useless at a 95% guarantee, its margin ranks pairs worse than plain confidence, and both analyses act almost only on NEI. Stronger verifiers are needed before support floors, weakest links or support-contextuality can be tested on real claims.

14.9 Claims and limits

Proposed in [41]. Support floors as ambient parameters; a presheaf of local probability, support and provenance data; weakest-link and typed-obstruction diagnostics; a context-carrying layer for history.

Immediate. A coupling polytope cut by finitely many support inequalities is a possibly empty polytope; the max–min program is linear; presheaves encode restriction without asserting a global section.

Established here. Cell floors are reduced margins (proposition 14.3), so the toric and chamber machinery applies unchanged and the weakest link has a closed form; the ambience is a functor into posets whose total category carries history and replay, policy imposition and floor summarisation are an idempotent comonad and monad on its fibres, and their interaction with transport is oplax (proposition 14.5; [34, Thm. 10.26] for whole chains); E3–E6 hold as pre-registered, E1 and E2 after documented corrections; support-contextuality occurs in a quarter of random gluable triangles.

Not established. Floors on events and on cyclic covers have no closed form and use linear programs; the weakest link is computed one source at a time; a falsehood shared by every source is not separable from the truth and shows only as a wide credal set; the AMB and Vigneaux cohomologies are not unified, and the non-additivity of the support indicator argues that no direct map exists; floors have not yet been tried in the recursive-training experiment, where they are a candidate repair for its failed prediction 3. On real claims (SciFact, section 14.8) the layer is right when it acts but, with weak classical verifiers, rarely acts.

15 Why the deeper machinery is not yet activated

Most of this paper uses the first layers of the programme: bands, regimes, Contextuality-by-Default and the gluing gap. The deeper machinery appears only as a description, or on synthetic data, or not at all:

- the three-level tower and its distributive law;
- Kleisli composition and whiskering;
- the square-zero cone;
- toric and Floer theory.

There are four reasons, and they are not the same for every layer (table 15).

1. *The datasets carry only the lowest layer.* A survey, a panel or a hospital table gives a few categorical variables in a few contexts. The deeper layers act on objects such data do not record:

    - candidate models with posteriors and a reliability level, for the tower;
    - a known channel applied to paired data, for whiskering and interchange;
    - randomized contexts that measure both contents on the same units, for contextuality.

    Without them the deeper operations have nothing to act on, or act on a single point, where they reduce to the lower layers.

2. *The questions asked were first-order.* Each analysis asked whether there is a tear, contextuality, a composition failure or a selection effect. None asked a question that only a deeper layer can answer, one where it predicts a number that the standard analysis gets wrong. A layer is activated by such a question, not by data alone. The interchange test of section 8.1 is the first: the theorem predicts that the monitor's composition null is wrong on banded data, and by how much.

3. *Several deeper layers collapse by the programme's own theorems.* Some layers are expected to add nothing on tables:

    - The gluing algebra is semisimple, $A_f \cong \mathbb{R} \times M_{n-1}(\mathbb{R})$, so every homotopy invariant of a functor out of it sees only the mass [3, Cor. 5.3].
    - The band monad identifies all parallel kernels at the level of 2-cells [7, Prop. 5.6].
    - The only strict distributive law is the base-relative one [34, Thm. 9.8], which is a product of free algebras.

    These layers settle which constructions are coherent. They do not produce new statistics, so it is expected, not a failure, that they are idle on contingency tables. Their information sits at chain level, in defects such as the whiskering and interchange defects, and those can be activated.

4. *Some machinery was used without its name.* The four-wave test composes a latent Markov chain with a measurement kernel, the setting of [9, Thm. 2.3]. Its lesson, that composition must be tested on latent states, is a statement about Kleisli composition with a stochastic kernel. The selection analysis asks whether three tables are margins of one law, which is a question about the fibres of the measurement map. Both were reported in statistical language.

Table 15: Which layers of the programme the data activate. Status: *active*, used as a test with a decision; *descriptive*, computed but predicting nothing; *implicit*, used under another name; *synthetic only*; *collapses*, adds nothing on tables by the programme's own theorem.

machinery	needs from the data	what our data supply	status	what would activate it
bands, regimes [7]	margins with sampling bands	every dataset	active	–
CbD, contextual fraction [8]	contexts measuring the same contents; randomized contexts for a causal reading	GSS split ballots (randomized); sites, units, recruitment waves (not randomized)	active, low power	designed experiments with both contents per respondent; larger samples (table 7)
holonomy of operations [33]	a cycle of operations on the same contents	the two GSS wordings over eleven items	active	more wordings, or prompt variants for language models

machinery	needs from the data	what our data supply	status	what would activate it
gluing, composition [9]	three or more waves on the same units	GSS, ICU, diabetes, C19PRC	active	–
latent composition	four or more waves	C19PRC	active, limited power	longer panels
Kleisli composition with a measurement kernel	a known or modelled channel	error kernels in the latent models	implicit	test–retest data, giving the kernel directly
interchange [3, Prop. 6.1]	a merge or kernel applied to paired data	banding of PHQ-9 totals and of ICU pressures	active (sections 7.2 and 8.1)	every banded analysis; recoding of survey answers
whiskering, multiplicativity defect [7, 9]	a stochastic channel acting on couplings	none known on our data	synthetic only	label noise with gold labels (crowd data); noisy relabelling of model outputs
three-level tower [34]	candidate models, their posteriors, a reliability level	none supplied; we fitted five models, with reliabilities and a family of floors, on PhysioNet 2012	active, replicated on new splits (sections 7.2 and 11.4)	new ICU data (eICU, MIMIC-IV); forecast and model ensembles
mixed distributive law, pooling	posteriors pooled across samples or sites	sites and units, but no site-level posteriors	synthetic only	multi-site pooling (heart-disease sites, ICU types) with per-site towers
selection as fibres	a measurement map with missingness	lactate, HbA1c, attrition	active: restricted models (section 11.2); blind to selection within (X, Y) cells	a fourth table, or an instrument, that sees Z directly
gluing algebra, square-zero cone [3]	– (structure of 2-cells)	–	collapses ([3, Cor. 5.3])	only its chain-level defects, through interchange and whiskering
toric surfaces, branes, entropy brane [8]	margins near walls or the boundary; an outcome to predict	real 2×3 and 3×3 margins; the tower's predicted margins; per-stay ICU systems	a diagnostic in the tower; forecasts GSS chamber changes; no forecast of ICU death (section 11.1)	nondegenerate per-unit systems (finer time grids), and a decided-tear target
Floer theory of couplings [8]	as above, plus a question about pairs of couplings	computed for small cases	descriptive	a comparison of couplings across contexts whose answer depends on Floer data
context-invariance variety $O \perp C \mid A$ (toric)	the same information in several contexts, and an outcome	GSS wordings × party; ICU units × pressure × death	active (section 12)	randomized contexts with outcomes in clinical data
boundary strata, families over bands	systems with empty categories	per-stay ICU day systems	active: geometry recovered, no forecast (section 12)	longer windows, finer grids

machinery	needs from the data	what our data supply	status	what would activate it
precision ladder [6]	sources with known supports	none	not used	attribution of a mixture to known sources (e.g. sites) with an unmodelled part

The pattern is clear. The layers that need only margins, waves or contexts are active. The layers that need a model space, a channel or a question about pairs of couplings are dormant on observational data. That is because the data do not supply those objects, and because we did not yet ask the questions that would need them. Several activations have now been made on data already in hand:

- the interchange correction, on the C19PRC and ICU bands (sections 7.2 and 8.1);
- a tower over PhysioNet with Fano tubes of its gradations, and a generalized reliability level that replicates (sections 7.2 and 11.4);
- geometry as a forecaster (section 11.1);
- the context-invariance variety and boundary-aware geometry (section 12);
- a family-of-cones tracker of the approach to collapse, and recursive training with known truth (section 13);
- ambient support geometry: a decision layer with support floors, typed obstructions and replayable records, carried by a fibred, locally posetal 2-category (sections 14 and 14.4);
- restricted selection models (section 11.2);
- a structural-event pipeline (section 11.3).

The next cheapest is label noise with gold labels in the crowd data of ThreeLayers.

16 What the test says about the theory

layer	prediction	observed
context (Layer 3)	change can be contextual or a direct influence; CbD separates them	no contextuality detected beyond direct influence in 1320 survey system-years, ICU types, admission sources, heart-disease sites or panel recruitment; the naive test is never zero; crime and drugs nearest the boundary, never decided; the test detects only large deficits at these sizes
positive control	the test detects contextuality where it exists	Snow Queen reproduced exactly by the cyclic criterion (0.452, interval $[0.226, 0.668]$); the canonical test holds its size, with about 50% power at deficits 0.08–0.16
selection	selective measurement makes available-case tables incompatible	incompatible beyond sampling: lactate (ICU, two sets), HbA1c (diabetes), attrition (UK panel); cell weighting restores compatibility by construction, so the mechanism is not tested further
operations (Layer 1)	a connection is flat iff its holonomy class vanishes	the wording connection on $K_{2,11}$ is non-flat in all 24 years
site separation	each layer's invariant is nontrivial on its own site and zero on the others (a structural remark)	consistent: Layer 3 not detected where Layers 1 and 2 are nonzero, in 340 system-years

layer	prediction	observed
toric	couplings are moment polytopes; branes avoid the product coupling	every real 2×3 surface computed; product carries no brane, leximin at least half; the survey tear is a blow-down at a fixed point, $15 \to 12$ vertices
probability	a summary is a margin; the coupling is a point of a polytope fixed by the margins	the same margins carry very different couplings: churn of panel answers and of damaged prompts
bending	small changes keep the chamber; stress < 1 certifies it	NHANES bends with certificate; ARC, LeetCode, OpinionQA bend with stress > 1
tearing	tears are wall crossings	three LLM tasks tear, all as rank reversals; one survey tear, not robust to multiplicity
breaking	an outcome reaches zero	none at the 1% level; one category emptied but not decided
degeneracy	a content at a vertex has a one-point polytope	OpinionQA: the largest change is a bend
composition	three contexts compose iff the gluing gap is zero	answers never glued (GSS, ICU, diabetes, C19PRC); with four waves a latent Markov chain is not rejected in either C19PRC strand, with limited power; three waves do not decide; diabetes regimens reject it at 5%, not at 1%
Kleisli and whiskering	kernels form a monad; whiskering is functorial and integral only for the stated classes	exact on a 15-kernel library; every stated failure reproduced with its size (section 10)
tower	three coherent levels; zeros break at level two; claims fail at a definite level	exact against Bayes on 200 steps; Cromwell break and planted claims at levels 1, 2, 3 detected where predicted; on PhysioNet, level three idle within one population and unable, as specified, to represent a shift, which its level-one certificate detects
distributive law	only the base-relative law satisfies all four axioms	reproduced; a planted law fails (M) by its closed-form amount; random instances can miss it
interchange	squares compose for bijections and sufficient merges only	reproduced on synthetic data; on the C19PRC panels banding is not sufficient, and the interchange defect carries 21–34% of the banded composition failure
outcomes	structure is associated with outcomes	states are associated with vote, party, death and self-harm thoughts; the order of states adds nothing detectable in any pre-registered test

Two lessons go back into the theory. First, the most consequential changes in these data are not tears of the margins but differences of couplings over nearly the same margins: churn under damaged information and failed composition. The programme's emphasis on the coupling, rather than the margins, is borne out. Second, couplings over the same units need a model of measurement before they can be read as change: gluing should be tested on the latent attitudes, not on the answers, and excess churn compared with what replicate answers would give. In the language of the programme, the measurement is a kernel from attitudes to answers, and composing a glued chain with a noisy kernel at each wave destroys gluing; the monitor's composition test belongs one level up, on the latent states.

The shortfalls of the programme, and the limitations of particular analyses, are collected in section 17.

17 Where the programme can do better

The data show where the theory holds. They also show where the programme, its theory, its monitor and its evidence, falls short. We list the shortfalls with what would remedy them, in order of importance.

1. *Composition must be tested on latent states, and the monitor should do it by default.*
    - The gluing gap of observed answers is positive in every panel and time series examined: GSS, PhysioNet 2012, Diabetes 130, and the UK and Irish C19PRC strands. Yet with four waves a latent Markov chain is not rejected in either C19PRC strand.
    - That non-rejection is weak evidence: the four-wave test detects a planted second-order memory in 3 of 8 replicates at $n = 771$ and in none at $n = 327$. Panels with more complete four-wave cases, or more waves, are needed for a decision.
    - As formulated in [7, Prop. 4.8], the gap measures unreliability as much as change.
    - Remedy, theory: state composition for hidden states, as the gluing of a latent chain pushed forward by a measurement kernel.
    - Remedy, monitor: report the latent test, with at least four waves, beside the observed gap.
    - The monitor's composition null is wrong on banded data. Banding is a non-sufficient merge, and by the interchange law it creates part of the failure: the correct null is 41–51% larger in the panels (section 8.1), and on planted first-order raw chains the banded null rejects every time. The ICU gaps, computed on banded pressures, were rechecked: the correct null is twice the monitor's, and the ICU failure survives it (section 7.2).
    - Three time points are too few. The latent Markov verdict flips between samples of the same population (PhysioNet sets A and B; UK and Irish wave 1–3). Two open cases remain: Diabetes 130, and the dense time series of MIMIC-IV.

2. *Contextuality is never detected outside the positive control, and the test is weak near the boundary.*
    - Across 1,320 survey system-years, eight retraining tasks, ICU types, admission sources, heart-disease sites and panel recruitment waves, no contextuality is detected beyond direct influence.
    - The positive control validates the cyclic criterion, not the canonical 3×3 test used for the non-detections. On planted systems that test holds its size, but detects deficits of 0.04 or less at most one time in ten and deficits of 0.08–0.16 about half the time (table 7). The non-detections exclude large contextuality only; the Irish wave-5 deficit of 0.040 ($p = 0.062$) is exactly the kind of effect the test would miss.
    - The boundary null is a natural noncontextual truth, not a proved least favorable one. The Dirichlet posterior screen is liberal near the boundary. Both use Kish effective sizes rather than the survey design.
    - Remedies: a more powerful test (a likelihood-ratio statistic in place of the deficit, contents with fewer categories when the question allows); power curves reported with every non-detection; design-based variances; designed experiments (randomized contexts measuring both contents on the same respondents) for systems such as crime and drugs that sit near the boundary.

3. *The selection analysis, as planned, could not fail.*
    - Weighting on the cells of the fully observed table makes the three tables compatible by construction (section 7), so the planned decision "selection explains the incompatibility" was certain whenever the available-case test rejected. What the data show is that measurement is not at random with respect to the measured states, not that a particular selection mechanism is the cause.
    - Done: restricted selection models with their own nulls (section 11.2), now also in the MIMIC-IV kit. They test only what a model implies about the (X, Y) composition of the measured. Selection within (X, Y) cells, including missingness not at random, remains untestable from three tables; a fourth table that sees Z directly is needed.

4. *Outcomes are associations, and the outcome claims are the weakest part of the evidence.*
    - The vote, party identification, death and thoughts of self-harm are associated with states; readmission in Diabetes 130 is not associated with the later regimens. No design here identifies a cause.
    - The UK outcome item asks whether the respondent has *ever* had thoughts of self-harm, and item 9 of the PHQ-9, part of the state, asks about such thoughts in the last two weeks. The association of the states with the outcome is partly built in; a PHQ-8 sensitivity analysis is needed.
    - The falsifiable clinical claim of [7, §11], that the tear margin or obstruction predicts deterioration beyond a severity score, has not yet been tested.
    - Order tests are confounded by measurement error, because the order of noisy states can carry information about latent counts. A latent-path version of analysis C is needed.

5. *No real break has been observed.*
    - The breaking regime has no empirical instance; the nearest is GPT-4's "no valid answer" emptying on 500 items, not decided.
    - Candidates: a medication withdrawn from the market (for example ranitidine in MIMIC-IV's last period), or a model that stops refusing a category altogether. Large samples are needed for the band to fall below the threshold.

6. *The toric and Floer layers forecast only chamber instability so far.*
    - The surfaces, branes and leximin couplings of real 2×3 systems are computed exactly. Branes of real 3×3 systems are not.
    - On survey margins the chamber, hence the surface, is rarely decided.
    - Tested now (section 11.1): the distance to the walls and a previous chamber change forecast the next point-estimate chamber change of GSS systems beyond the margins. Per-stay ICU geometry forecasts neither death nor shock, because those systems are almost all broken. A forecast of decided tears, and of outcomes from nondegenerate systems, is still missing.

7. *Noise floors are missing where they matter most.*
    - Excess churn is an upper bound on masked change until replicate measurements give a floor. The LLM data have none (the USMLE "replicates" are damaged prompts), and the panels' within-wave reliability is unknown.
    - Remedies: repeated sampling of models at fixed settings, and test–retest designs.

8. *Pre-registration came late.* The LLM and GSS analyses (sections 4 to 6) were developed on the data and revised after independent reviews. Only the analyses of section 7 after NHANES and

of section 8 were pre-registered and replicated, and even those have six deviations from their plans (section 7.6), two of them found only after freezing. The earlier findings should be read as exploratory until replicated, for example on later GSS years or other model versions.

9. *Design and weighting.*
    - Panels and clinical data are analysed unweighted and on complete cases. The six-wave analysis B3 uses 582 of the 2,025 UK wave-1 respondents, a selected group.
    - Inverse-probability weighting uses only cell covariates, so data missing not at random are not handled.
    - The open ICU data lack vasopressors, so shock is defined by pressure alone.
    - Encounter order in Diabetes 130 is assumed.
10. *Random-instance checks have blind spots, and the structural tests are finite.*
    - The programme's algebraic claims are checked by scripts that evaluate each law on random instances. In T2 a planted law that fails the multiplication axiom passed every random instance at one value of its parameter, because the generator draws only two-point samples (section 10). A passing random check is evidence only against failures its generator can produce.
    - Remedy: generate instances adversarially, varying support sizes and degenerate cases, and keep a targeted counterexample for every stated failure, as T1–T4 now do.
    - The synthetic tests confirm the code against the papers on finite cases. They do not replace the proofs, and they do not show that the monad, the tower or the interchange law carry information in data. The interchange law and the tower have now been run on data (sections 7.2 and 8.1).
11. *The tower's reliability level was too narrow, and the toric diagnostic needs its own bands.*
    - Done: a family of floors, a posterior over binary channels, earns its keep in all three new shift arms, and the evidence floor in two (section 11.4). Open: replication on new ICU data, and a reliability kernel over more than two outcomes.
    - Linear discounting towards a fixed floor can only shrink a model towards that floor. It cannot represent a shift that raises an outcome's prevalence, which is the commonest transport failure in clinical data. In the PhysioNet shift arm, level three moved most of its mass below full reliability but could not improve the forecasts. The lifting certificate showed why: the observed rate lies outside every admissible prediction (section 7.2).
    - Remedy: let level three carry a family of floors, or a reliability kernel rather than a scalar. The repaired floor was chosen after seeing the result, so it must be pre-registered and replicated before it counts.
    - The tower's posterior over toric surfaces ignores the sampling uncertainty of the realized margins. It was certain of a surface that the data's own bands leave undecided. Remedy: propagate the bands of the predicted and observed margins into the surface posterior.
12. *Scale and engineering.*
    - The canonical CbD program grows exponentially with the number of outcomes, so the monitor bands every content into three categories, which loses information.
    - The latent models fix three classes, with no selection of their number. Their degrees of freedom are nominal, since some parameters sit on the boundary, which is why the four-wave test uses a parametric bootstrap. The three-wave tests still use the χ^2 reference.
    - The monitor is a set of checked scripts, not a tested package.

Further limitations of particular analyses.

- No multiplicity correction is applied across the 138 survey steps; under Bonferroni the one survey tear is only a possible tear.
- LLM answers are single samples at temperature 0.1, parsed from free text by regular expressions; parsing moves accuracies by about one point.
- The NHANES diabetes variable is self-reported, and only two cycles are used.
- Toric surfaces and branes are computed for margins rounded to three decimals; where a chamber is not decided on the bands (fig. 6), the surface is a point estimate.
- The C19PRC outcome item asks about thoughts of self-harm *ever* and overlaps PHQ-9 item 9; the Irish outcome had too few positive answers (8) to be analysed.
- The Irish wave-5 recontact flag is constructed (section 7.6); a respondent who missed wave 1 but joined at wave 2 counts as fresh at wave 5.

18 Datasets of the programme, and what is new here

Table 17 lists every dataset used in the ten papers of the programme, with what it is used for. Real data are analysed in ThreeLayers, Bands, Tower and this paper; the other papers use synthetic data or worked examples, and some quote ThreeLayers' real-data results (Conditionals, for example, its MBIC contextual fractions). All code is in the repository `https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent` under the folder named for each paper.

Table 17: Datasets of the programme. Type: public (real, openly available), credentialed (real, under a data use agreement), synthetic (generated by the papers' scripts), example (worked by hand).

paper	dataset	type	used for	source
Validation (this paper)	LLM drift: GPT-3.5 and GPT-4, March and June 2023, eight tasks	public	regimes, contextuality, churn under retraining and prompt damage	[10], `github.com/lchen001/LLMDrift`
	General Social Survey 1985–2024, eleven spending items in two wordings (55,169 respondents for welfare)	public	site separation, tears over 40 years, toric types, geometry as a forecaster of chamber changes, structural events against partisanship	[19] via `gssr` [20]
	GSS panels 2006–10, 2008–12, 2010–14	public	churn, gluing, latent models, path dependence, vote and party	[19] via `gssr`
	NHANES 2009–10 and 2011–12	public	negative control, toric surface of a clinical system	[27]
	Snow Queen experiment, 1,799 respondents (published tables)	public	positive control for contextuality	[14]
	planted two-context 3×3 systems	synthetic	size and power of canonical CbD	`ext/snowqueen/`
	PhysioNet/CinC Challenge 2012, sets A and B, $2 \times 4{,}000$ ICU stays	public (ODC-By)	pre-registered analyses A–E, replication, negative control; interchange recheck; three-level tower, Fano tubes and generalized reliability; per-stay geometry as a forecaster; restricted selection	[28, 18]

paper	dataset	type	used for	source
	Diabetes 130-US Hospitals, 101,766 encounters	public (CC BY 4.0)	selective HbA1c measurement, insulin paths, readmission (not associated)	[32, 35]
	Heart Disease, four sites, 920 patients	public (CC BY 4.0)	contextuality across sites, missingness (illustration)	[12, 36]
	C19PRC-UK waves 1–6, 2,025 at wave 1	public (OSF)	four-wave latent composition, panel conditioning, attrition, banding interchange test, restricted selection models	[25], `https://osf.io/v2zur`
	C19PRC Ireland (Mental Health ASAP) waves 1–5, 1,041 at wave 1	public (OSF)	replication of the C19PRC-UK analyses	[31]
	MIMIC-IV	credentialed	not analysed; pre-registered kit tested on synthetic data	[26]
	SciFact, 1,409 claims and 5,183 abstracts (340 dev pairs)	public (CC BY-NC 2.0), kept outside the code archive	decision layer on real claims: credal channels, conformal reliabilities, guarantee and coverage	[49]
	T1–T4: kernel library, tower with a Cromwell model and reliabilities (200 observations), contexts, couplings	synthetic, frozen	structural backbone with planted failures	`ext/structure/`
	synthetic data in the layouts of PhysioNet 2012, Diabetes 130, Heart Disease, C19PRC and MIMIC-IV, with planted answers	synthetic	tests of every adapter before the real data	`ext/`
Bands	NHANES 2009–10 and 2011–12, 10,730 adults, 16 subgroups	public	information ladder, out-of-sample bands, barcodes	[27]
	US COVID-19 Forecast Hub ensemble, 56 jurisdictions, 9,072 forecasts	public	calibration, tear time, unreliable data	[11]
	MIMIC-IV	credentialed	protocol only	[26]
	planted checks S1–S3	synthetic	validation of tests	`bands/`
ThreeLayers	BASIL, 100 events	public	contradiction meter, three-level bias, dynamics	Fan et al., EMNLP 2019
	MBIC, 1,700 sentences, 17,775 labels	public	bounds, grey contexts, toric sphere	Spinde et al., iConference 2021
	NewsWCL50, 50 articles	public	hull cover (the corpus that tears)	`github.com/fhamborg/NewsWCL50`
	AllSides, 37,554 articles 2012–2020	public	events, centre–right shift, streams	`github.com/ramybaly/Article-Bias-Prediction`
	BABE, 1,700 and 3,674 sentences	public	hull representation	Spinde et al., Findings of EMNLP 2021
	FiveThirtyEight approval polls, 5,920	public	outcome series, certificate	CC BY 4.0, FiveThirtyEight

paper	dataset	type	used for	source
	2016 national polls, 562	public	curvature, singular strata	`github.com/TheEconomist/us-potus-model`
	CrowdER, 8,315 pairs	public	crowd covers	Wang et al., PVLDB 2012
	JetClass-II, 300,000 jets sampled	public (simulated physics)	second-order information geometry, holonomy	`huggingface.co/datasets/jet-universe/jetclass2`
	UD English-EWT, 16,622 sentences	public	monodromy	`github.com/UniversalDependencies`
	CausalIDView, 40 worlds	public benchmark	acquisition, support and Bayesian priors	Jung et al. 2026
	corpora A, B, V; O6 benchmark; planted versions of BASIL, AllSides, crowd and jet data	synthetic	calibration of every test	`AMB-Vigneaux/`
Tower	CausalIDView, 40 worlds	public benchmark	Manski and IV bounds	Jung et al. 2026
	deep-ensemble and label-noise data, nine CSV files	synthetic	hallucination tests, label noise	`synthetic_examples.py`; archive `coupling/synthetic_data.zip`
Conditionals	synthetic moderation data, $N = 2000$	synthetic	tests of §12, chambers, horns	same script and archive
Companion	the same synthetic files (Y1–Y8)	synthetic	tabulated in its §15	same script and archive
Algebras	first 40 items of the moderation data	synthetic	Fisher's exact test	same archive
Attribution	four-stage provenance history	synthetic	attribution demo	`provenance_demo.py`
Bridges	three bits; boundary detection (2000 replicates)	synthetic	brane boundary test	`check_bridges.py`
	real margins of this paper	public	toric surfaces and branes (section 9)	this paper
Architecture	$f = (0.4, 0.6)$, $g = (0.3, 0.43, 0.27)$	example	one example through every layer	`architecture/`

The synthetic files are archived at `https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/blob/main/coupling/synthetic_data.zip`.

What is new in this paper.

- *A monitor of structural change*: bands, the regime of each change, stress against the tear margin, canonical Contextuality-by-Default, excess churn over the maximal coupling, the gluing gap and latent measurement models, in one library with checks.
- *An empirical separation of sites*: on the same survey, contextuality (Layer 3) is absorbed by direct influence, while the holonomy class of the wording connection (Layer 1) is nonzero every year and the items are dependent (Layer 2); a test with identical connections confuses Layer 1 with Layer 3.
- *A boundary-null test of canonical contextuality*, combined over years and corrected over pairs, which is calibrated where a posterior screen is not.
- *Toric geometry and Floer theory on public margins*: the toric surface of every real 2×3 system, the exact nonzero torus branes, the leximin and product couplings; tears as blow-downs of the

3 × 3 polytope; the entropy brane and its steepness as a measure of closeness to breaking.

- *Panels*: composition fails for answers but a latent Markov chain is not rejected; the order of answers adds nothing to their association with the vote or party beyond the answers themselves.
- *A positive control and a power check*: the Snow Queen contextuality is reproduced exactly; the canonical test holds its size on planted systems but detects only large deficits at the sample sizes here.
- *Pre-registered clinical replications*: available-case tables of selectively measured quantities are incompatible beyond sampling in two ICU samples, in diabetes and in a panel; within severity strata no contextuality is detected between units or sites.
- *A four-wave latent Markov test*: in two countries, answers that do not compose are consistent with a latent first-order Markov chain seen with error; the test's power against longer memory at these sizes is reported.
- *An illustration* on four heart-disease sites, which differ mainly in what they measured.
- *The structural backbone on synthetic data*: Kleisli composition, the three-level tower, its mixed distributive law and the interchange law, each with closed-form ground truth and planted failures of known size, pre-registered and hash-frozen.
- *Retraining*: tears are rank reversals of the two models; damaged prompts change which items are right far more than how many; the USMLE data contain a damaged copy of every prompt.

19 Computational checks

All code is at `https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/tree/main/monitor`. `monitor.py` implements the monitor and its unit tests; `llm_analysis.py`, `usmle_prompts.py`, `gss_analysis.py`, `clinical_analysis.py`, `cbd_boot.py` and `panel_models.py` produce the results of sections 4, 5 and 7; `site_separation.py`, `boundary_tests.py`, `ctx_allpairs.py` and `panel_paths.py` those of section 6; `toric_public.py`, which calls the scripts of `bridges/fukaya`, those of section 9. `make_tables.py` and `make_figures.py` produce the tables and figures from the saved results. The folder `ext/` holds the pre-registered work of sections 6.2, 7 and 8: the plans (`PREANALYSIS_*.md`), the adapters (`open_clinical/challenge2012.py`, `diabetes130.py`, `heart.py`; `c19/c19.py` with the four-wave model `latentT.py`; `snowqueen/snowqueen.py`), their synthetic tests with planted answers, the frozen results with their hashes, the post-review checks (`c19/b2_equal_starts.py`, the four-wave rerun and its power; `snowqueen/cbd_split_power.py`, the planted check of canonical CbD), the tower and Fano tubes on PhysioNet and the ICU interchange recheck (`open_clinical/tower_icu.py`, with its exploratory follow-up `tower_icu_explore.py`; figure by `make_fig_fano.py`), the banding interchange test (`c19/banding_interchange.py`), phase 4 (`phase4/`: plan, `geom.py`, `wp1_clinical.py`, `wp1_gss.py`, `wp2_selection.py`, `structural_events.py`, `wp4_tower.py`, their planted checks and frozen results; figure by `make_fig_phase4.py`), phase 6 (`phase6/`: `fan_tracker.py`, `trajectories.py`, `recursive_training.py`, plan and frozen results; figures by `make_fig_collapse.py`), phase 7 (`phase7/`: `decide.py`, `support.py`, `experiments.py`, `rechecks.py`, plan and frozen results), phase 5 (`phase5/`: `context_map.py`, `boundary_geometry.py`, plan and frozen results; figure by `make_fig_context.py`), the structural tests T1–T4 (`structure/`, with their plan, frozen data and frozen results), and the MIMIC-IV kit (`mimic_kit/`). `check_monitor.py` reruns the unit tests and the fast analyses, verifies the saved outputs of the slow ones, checks propositions 2.2 and 2.4 on every reported change, and pins every number quoted in the text. The data are public: the LLM-drift answers from the authors' repository, the GSS through `gssr`, and NHANES through the `NHANES` R package.

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The data are used under their licences: CC BY 4.0 for the UCI data, Open Data Commons Attribution for the 2012 Challenge, and the terms of the GSS, NHANES, LLM-drift and C19PRC releases. No credentialed data were accessed for this paper. Independent reviews by language-model referees found several errors in earlier drafts, which are corrected here. The author is responsible for all remaining errors.

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